Lorem Ipsum

Student Name: Lorem Ipsum
Roll Number: 202xxxx

BTP report submitted in partial fulfilment of the requirements
for the Degree of B.Tech. in Computer Science & Engineering

on November 11, 2026

BTP Track: Research

BTP Advisor
Prof. Lorem Ipsum

Indraprastha Institute of Information Technology
New Delhi

Student Declaration

I hereby declare that the work presented in the report entitled **"Lorem Ipsum"** submitted by me for the partial fulfilment of the requirements for the degree of *B.Tech. in Computer Science & Engineering* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of my work carried out under the guidance of **Prof. Lorem Ipsum** . Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

Lorem Ipsum **Place & Date:** November 11, 2026

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Prof. Lorem Ipsum **Place & Date:** November 11, 2026

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Keywords: (e.g.Mobile computing, security, complexity, algorithms, image analysis, machine learning, information retrieval)

Acknowledgment

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Work Distribution

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Contents

Student Declaration i

Abstract ii

Acknowledgment iii

Table of Contents vi

1 Introduction 1

1.1 Background . . . 1
1.2 Digital Twins in Medicine and Nutrition . . . 2
1.3 Computational Gastronomy and Recipe Data . . . 2
1.4 Problem Statement and Research Gap . . . 3
1.5 Research Question and Objectives . . . 3
1.6 Approach Overview . . . 4
1.7 Contributions . . . 4
1.8 Scope and Limitations . . . 5
1.9 Thesis Outline . . . 6

2 Literature Review 7

2.1 MASLD and Its Dietary Management . . . 7
2.1.1 Definition and burden . . . 7
2.1.2 Evidence for lifestyle and dietary treatment . . . 7
2.2 Non-Invasive Assessment of Liver Fat . . . 8
2.3 Machine-Learning Prediction of Steatosis and Liver Fat . . . 9
2.3.1 Classification of steatosis in NHANES . . . 9
2.3.2 Prediction of liver fat as a continuous value . . . 10
2.3.3 Why prediction from routine data is bounded . . . 10
2.4 Dietary Assessment and Personalised Nutrition . . . 11
2.5 Digital Twins in Health and Nutrition . . . 12
2.5.1 Definitions and requirements . . . 12

2.5.2 Metabolic and nutrition twins . . . 12
2.5.3 From prediction to "what if" . . . 13
2.6 Computational Gastronomy and Food Recommendation . . . 14
2.6.1 Recipe and food resources . . . 14
2.6.2 Food recommender systems . . . 14
2.6.3 Recipe modification and adaptation . . . 14
2.6.4 Menu planning, diversity and additional criteria . . . 15
2.7 Evaluating Digital Twins Without Longitudinal Data . . . 15
2.8 Summary and Research Gap . . . 16

3 Experiments on Predicting Liver Fat **18**
3.1 Evaluation Protocols . . . 18
3.2 Experiments . . . 18
3.2.1 Phases 1 and 2: baselines, tuning and inputs . . . 19
3.2.2 Phase 3: features, weighting and learners . . . 19
3.2.3 Phase 4: the target and other model families . . . 20
3.2.4 Phase 5: additional data . . . 20
3.2.5 Phase 6: target population and lean participants . . . 21
3.3 Final Model and Results . . . 21
3.4 Why R^2 Plateaus . . . 23
3.5 Comparison with Published Models . . . 25
3.6 Summary . . . 26

4 Methodology **27**
4.1 Data . . . 27
4.1.1 NHANES FibroScan cohort . . . 27
4.1.2 Habitual diet from the 24-hour recall . . . 28
4.1.3 Recipe corpus . . . 28
4.1.4 NHANES III: external cohort with long-term follow-up . . . 28
4.2 Layer 2: Liver-State Model and Clinical Safety Checks . . . 29
4.2.1 Liver-state model . . . 29
4.2.2 Uncertainty and explanation . . . 29
4.2.3 Fibrosis risk, liver label and safety notices . . . 29
4.3 Layer 3: Evidence Engine . . . 30
4.3.1 Intervention evidence table . . . 30
4.3.2 From patient state to dietary priorities . . . 30
4.3.3 Causal knowledge graph . . . 30
4.4 Layer 4: Recipe Evaluation and Ranking Engine . . . 31

4.4.1 Food preferences . . . 31
4.4.2 Criteria . . . 31
4.4.3 Health-gated ranking . . . 32
4.4.4 Explanations, diversity and weekly plans . . . 33
4.5 Layer 5: Scenario Engine . . . 33
4.6 Updating the Twin: Plan, Check-ins and Re-assessment . . . 34
4.6.1 The plan . . . 34
4.6.2 Predict, observe, compare, adjust . . . 34
4.6.3 Tips and alerts . . . 35
4.7 Plain-Language Summary . . . 35
4.8 System and Interfaces . . . 35
4.9 Evaluation Design . . . 36
4.9.1 Liver-state model and safety checks . . . 36
4.9.2 Fairness audit . . . 36
4.9.3 External and long-term validation on NHANES III . . . 36
4.9.4 Recipe ranking . . . 37
4.9.5 Plain-language summaries . . . 37
4.9.6 In-silico evaluation of the updating twin . . . 37

5 Results **39**

Bibliography **40**

Chapter 1

Introduction

1.1 Background

Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly non-alcoholic fatty liver disease (NAFLD), is the most common chronic liver disease worldwide. It affects about 30% of the adult population, and its prevalence has risen steadily over the past three decades [1], [2]. In 2023 a multisociety consensus renamed the condition to reflect its metabolic origin: steatotic liver disease accompanied by at least one cardiometabolic risk factor, such as excess weight, raised blood glucose or dyslipidaemia [3]. Untreated, MASLD can progress from simple fat accumulation (steatosis) to inflammation, fibrosis and cirrhosis, and it is closely linked to type 2 diabetes and cardiovascular disease [4].

Diet is central to its management. No dietary pattern is a cure, but the current European guideline recommends lifestyle change, centred on diet quality and weight loss, as the first-line treatment for most patients [4]. The evidence is graded by the amount of weight lost: in a prospective study with paired liver biopsies, a loss of at least 5% of body weight reduced steatosis, 7–10% improved inflammation, and 10% or more led to resolution of steatohepatitis in most patients and to regression of fibrosis in many [5]. Beyond weight, the guideline advises a Mediterranean-style diet, fewer foods rich in saturated fat and sugars, and the avoidance of sugar-sweetened beverages [4]. Trials and meta-analyses support parts of this advice, although the evidence differs by outcome: Mediterranean and calorie-restricted diets improve some liver measures [6], [7], Mediterranean and low-fat diets do not clearly differ [8], and saturated fat and fructose are linked to liver fat in feeding studies [9], [10].

Turning this guidance into daily meals is where patients struggle. Guidelines speak in nutrients and dietary patterns ("limit saturated fat", "increase fibre"), whereas people decide what to eat one recipe at a time, constrained by the cuisine they know, the ingredients they have, the time they can spend cooking and what they enjoy. The same advice also does not suit every patient equally: the priority for someone whose diet is already high in fibre but rich in saturated fat differs from that of someone with the opposite pattern. Bridging the gap between population-level evidence and an individual patient's plate is the problem this thesis addresses.

Liver fat itself can be measured non-invasively. Vibration-controlled transient elastography (FibroScan) reports the controlled attenuation parameter (CAP), which grades steatosis with established cut-offs [11], [12]. The US National Health and Nutrition Examination Survey (NHANES) performed this examination on a nationally representative sample in its 2017–March 2020 and 2021–2023 cycles, together with blood tests, body measurements, questionnaires and a 24-hour dietary recall [13]. In the 15,300 participants with a complete examination used in this work, 52% had a CAP of at least 248 dB/m, the usual threshold for any steatosis [11].

1.2 Digital Twins in Medicine and Nutrition

A digital twin is a virtual representation of a specific physical system that is updated with data from that system, can predict how it will behave, and is used to inform decisions about it [14]. In medicine the physical system is a patient, or one of their organs. Medical digital twins are most advanced in cardiology and oncology, and their development faces common challenges in data integration, validation and privacy [15], [16]. What distinguishes a twin from a one-off prediction model is that it is specific to one patient, is updated as new data arrive, and can simulate how that patient's state may change under different interventions [14], [16]. Systematic reviews report growing interest in digital twins for metabolic and population health, but also limited evidence of their effectiveness [17], [18].

Nutrition is a natural application. People respond differently to the same food [19], and the "virtual digital twin" concept in precision nutrition proposes combining a person's clinical, lifestyle and dietary data into a model that can test dietary choices before they are made [20]. Early clinical results are encouraging: in a one-year randomised trial in people with type 2 diabetes, nutrition guided by a digital twin that predicted each person's post-meal glucose response improved markers of liver fat [21]; related work applies the same technology to glycaemic control and diabetes remission [22], and a recent student abstract proposes twins of diet-influenced biomarker trajectories [23]. In MASLD specifically, the European ARTEMIS project is building virtual twins from retrospective data on several thousand patients [24]. None of these systems, however, evaluates complete, existing recipes against a patient's evidence-based dietary needs.

1.3 Computational Gastronomy and Recipe Data

Computational gastronomy treats food and cooking as data that can be analysed and reasoned about [25]. Structured resources now make this possible at scale. RecipeDB contains more than 118,000 recipes from 74 countries, each with its ingredients, cooking processes, utensils, cuisine and nutritional profile [26]; FlavorDB describes the flavour molecules found in foods [27]; and literature-mined resources link culinary herbs and spices to health outcomes [28]. Related work in menu planning uses multi-objective optimisation to compose daily menus that meet nutritional targets [29], and health-aware food recommenders have used food knowledge graphs to match recipes to nutritional needs [30], [31], [32], [33]. Another line of work generates new recipes or modifies existing

ones, for example by substituting ingredients or adapting a dish to another cuisine [34], [35], [36], [37].

These resources are designed mainly for exploring and querying culinary data, for generating variants, or for meeting generic nutrient targets. They do not, by themselves, decide which existing recipes suit a particular patient, given that patient's current liver state, the strength of the clinical evidence behind each dietary change, and the practical realities of their kitchen.

1.4 Problem Statement and Research Gap

Three bodies of work hold complementary pieces of the problem. Clinical evidence links dietary changes to liver outcomes; patient-specific models, including digital twins, give an estimate for an individual's health state; and recipe databases describe the food people actually cook. These components are usually studied separately. In particular, we found no framework that evaluates and ranks existing recipes for an individual patient by combining

- **health relevance**, derived from the patient's current liver and metabolic state and from graded intervention evidence;
- **cuisine compatibility**, reflecting the patient's culinary context; and
- **practical feasibility**, reflecting cooking time, number of ingredients and complexity,

within a transparent framework in which every recommendation can be traced back to the evidence that supports it.

1.5 Research Question and Objectives

This thesis asks:

> *Can a patient-specific liver digital twin, built from clinical data and graded intervention evidence, support the evidence-grounded evaluation and ranking of existing recipes according to their health relevance, cuisine compatibility and cooking feasibility for an individual patient?*

To answer it, the work pursues five objectives:

1. Estimate each patient's current liver state, as calibrated probabilities of fatty liver, from routinely collected clinical data.
2. Translate that state, together with the patient's own reported diet, into evidence-graded nutritional priorities, without inventing individual numeric targets.
3. Evaluate and rank existing recipes from RecipeDB against those priorities, the patient's cuisine and practical constraints.

4. Compare recipe choices as scenarios whose projected effects are grounded in clinical evidence, and update the patient's state as new data arrive.

5. Evaluate each component against appropriate baselines and state clearly what has and has not been validated.

1.6 Approach Overview

The proposed system, HepatoTwin, is organised in seven layers (Figure 1.1). Layer 1 holds the patient's state: demographics, body measurements, blood tests, questionnaire answers and a 24-hour dietary recall. Layer 2, the core of the twin, estimates the current liver state with an XGBoost model [38] and two calibrated classifiers [39] that give the probability of CAP at or above 248 and 280 dB/m, explained per feature with SHAP values [40]. Layer 3, the evidence engine, converts this state into nutritional priorities such as weight management or limiting saturated fat; each priority cites rows of a curated table of 23 intervention–outcome findings drawn from trials, meta-analyses and guidelines [4], [5], [6], [7], [8], [41], [42] with explicit evidence-strength labels, and is adjusted by the patient's own intake. Because a single 24-hour recall is a noisy measure of habitual diet [43], [44], [45], it can shift a priority by at most one level. Layer 4, the recipe evaluation and ranking engine (NutriTwin), is the core research contribution: it scores candidate recipes for health relevance, cuisine compatibility, practical feasibility and acceptability, ranks first the recipes that meet every evidence-linked reference value, and explains each recommendation. Layer 5 compares recipe scenarios and projects their effect over time. A three-dimensional liver and an explainable dashboard are used in Layers 6 and 7 for visualisation. The twin updates its state and its recommendations accordingly when new food logs or measurements arrive, in the spirit of assimilating new observations into a model state [14], [46].

The twin is built and evaluated on NHANES. Its liver-state model is intended for patients aged 12 to 59 and is evaluated against reliable FibroScan examinations; ages 60 and over are outside its scope.

1.7 Contributions

The main contributions of this thesis are:

1. **A calibrated liver-state model from routine data.** Trained on both NHANES FibroScan cycles, the model detects steatosis with ROC-AUC 0.881 (CAP $\geq$ 248 dB/m) and 0.879 (CAP $\geq$ 280 dB/m) on held-out patients within its scope, above the Fatty Liver Index (0.861) and the Hepatic Steatosis Index (0.836) on the same patients [47], [48], and holds its performance on a later survey cycle. Its discrimination is in the range reported by other machine-learning models built on NHANES routine data [49], [50], [51], [52]. It also predicts CAP with a held-out R^2 of 0.56, comparable to or above published CAP-prediction models built on routine data [53].

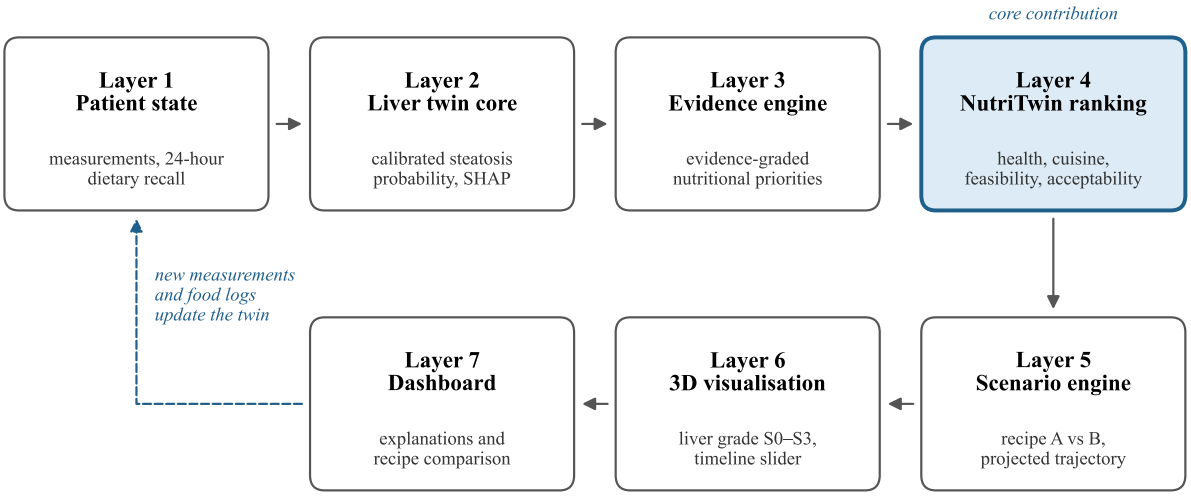


Figure 1.1: Overview of HepatoTwin. Patient data (Layer 1) calibrate the liver twin core (Layer 2), which guides evidence-graded priorities (Layer 3) and the ranking of existing recipes (Layer 4, the core contribution). Scenarios (Layer 5) are shown through a visualisation (Layer 6) and a dashboard (Layer 7); new measurements and food logs update the twin.

2. **An evidence engine linking patient state to graded dietary priorities.** Every priority cites named trials, meta-analyses or guidelines with an explicit strength label, observational findings are recorded but never used as effects, and the patient's own 24-hour recall personalises the priorities.

3. **NutriTwin, an evidence-gated multi-criteria recipe ranking engine.** On 300 held-out patients, 83% of its top-ten recipes met every evidence-linked reference value, the patient's cuisine and the practical limits at once, against 73% for weighted scoring without the health gate and under 2% for health-only ranking; it was the only method whose recommendations always met the nutritional references.

4. **Scenario comparison and weekly meal plans** that respect the patient's cuisine and nutritional references, with projected effects labelled by the strength of their evidence.

5. **An open, modular digital-twin framework**, demonstrated for MASLD, together with an in-silico evaluation design that tests how the twin updates on virtual patients before real follow-up data exist, following the use of simulated patients and virtual trials in other fields [54], [55], [56], [57].

1.8 Scope and Limitations

NHANES is cross-sectional: each participant was examined once. The twin's estimate of the current state is therefore validated on held-out patients, but its projections of future states cannot be compared with real repeat measurements. They are presented as simulated projections grounded in clinical evidence, never as clinical forecasts, and the twin's updating is evaluated on virtual patients

[54], [57]. For validation with real longitudinal data, deployment with patients who enter their own measurements over time is required. The recommendations are research outputs: they support, and do not replace, clinical care.

1.9 Thesis Outline

Chapter 2 reviews related work on MASLD and its dietary treatment, non-invasive liver fat assessment, machine-learning prediction of steatosis, dietary assessment, digital twins, computational gastronomy and the evaluation of twins without longitudinal data. Chapter 3 describes the methodology: the NHANES cohort and the recipe corpus, the liver-state model, the evidence engine, the recipe evaluation and ranking engine, and the scenario and updating components of the twin. Chapter 4 reports the experiments carried out to improve the prediction of CAP, and explains why its performance reaches a plateau. Chapter 5 presents the results of each component against its baselines.

Chapter 2

Literature Review

This chapter reviews the work on which HepatoTwin builds and the gaps it addresses. It follows the structure of the system: the clinical problem and the evidence for dietary treatment (Section 2.1); how liver fat is measured without biopsy (Section 2.2); machine-learning models that estimate steatosis from routine data, and the limits on how well they can do so (Section 2.3); how diet is measured and personalised (Section 2.4); digital twins in health and nutrition (Section 2.5); computational gastronomy and food recommendation (Section 2.6); and how a twin can be evaluated when no longitudinal data are available (Section 2.7). Section 2.8 summarises the gap.

2.1 MASLD and Its Dietary Management

2.1.1 Definition and burden

Metabolic dysfunction-associated steatotic liver disease (MASLD) is defined as hepatic steatosis together with at least one of five cardiometabolic risk factors, in the absence of another cause [3]. The 2023 multi-society nomenclature replaced non-alcoholic fatty liver disease (NAFLD), removed the reliance on exclusion and stigmatising terms, and introduced MetALD for patients with MASLD who also drink moderate amounts of alcohol. A meta-analysis of studies which used imaging to diagnose NAFLD estimated a global adult prevalence of 30.05%, rising from 25.26% in 1990–2006 to 38.00% in 2016–2019 [1]; a second meta-analysis reached a similar overall estimate [2]. Most patients have simple steatosis, but a minority progress to steatohepatitis, fibrosis and cirrhosis, and the disease is closely associated with type 2 diabetes and cardiovascular risk [4], [58].

2.1.2 Evidence for lifestyle and dietary treatment

Both the European (EASL-EASD-EASO) and American (AASLD) guidelines name lifestyle change as the first-line treatment for most patients [4], [58]. The strongest evidence is about weight loss. In a prospective study of 293 patients with paired liver biopsies, weight loss through lifestyle change produced dose-dependent histological improvement: losses of 5% or more improved steatosis and

inflammation, and losses of 10% or more resolved steatohepatitis in 90% of patients and led to fibrosis regression in 45% [5]. A meta-analysis of 22 randomised trials with 2,588 participants found that weight-loss interventions lowered ALT by 9.81 U/L and improved liver fat and histological activity, but found no significant change in fibrosis [59]. The thresholds of 5%, 7–10% and 10% weight loss are now written into the European guideline [4]. Liver fat also responds quickly: in people with type 2 diabetes on an 800 kcal per day diet, hepatic triglyceride fell by 30% after the first week, and eventually dropped from 12.8% to 2.9% over eight weeks [60].

The evidence for specific dietary components is more general, and this variation is important for any system that turns evidence into advice. A 2026 meta-analysis of randomised trials, which graded the certainty of each finding with GRADE, found that a Mediterranean-style diet reduced MRI-PDFF by 1.37 percentage points and improved liver stiffness, while omega-3 supplementation and low-carbohydrate high-fat diets did not change ALT [42]. Dietary interventions including calorie restriction improved liver enzymes, steatosis and stiffness in another meta-analysis [7]. Yet a meta-analysis of ten trials found that the Mediterranean diet reduced AST and GGT but not ALT [6], and it was no better than a low-fat diet in the short term [8].

Feeding trials show that the type of surplus energy matters. When 38 overweight adults were overfed by 1,000 kcal per day for three weeks, liver fat rose by 55% with saturated fat, 33% with simple sugars and 15% with unsaturated fat [9]. A meta-analysis of 13 trials with 260 participants found that fructose raised liver fat and ALT when it added surplus energy, but had no effect when it replaced other carbohydrate in the same amount of energy [10]. Exercise also reduces liver fat: in a meta-analysis of trials measuring liver fat by MRI, exercise was associated with a relative reduction of 30% or more, the threshold that predicts histological response, even without clinically significant weight loss [41].

Two results follow. The available evidence varies in strength, with biopsy-based dose–response findings on weight loss generally providing more direct support than meta-analyses based on liver enzyme changes. The literature also includes studies reporting no significant effects alongside those showing positive outcomes. Food recommendation systems should therefore communicate the quality and certainty of the supporting evidence for each recommendation instead of assigning equal weight to all findings. Second, the evidence is expressed in terms of energy, weight change, nutrients and dietary patterns, never single recipes; any mapping from evidence to a recipe is an inference that must be made explicit.

2.2 Non-Invasive Assessment of Liver Fat

Liver biopsy is the reference standard for grading steatosis and fibrosis but is invasive and impractical for screening. Magnetic resonance imaging proton-density fat fraction (MRI-PDFF) and proton spectroscopy (^{1}H-MRS) measure liver fat accurately but are costly. Vibration-controlled transient elastography (FibroScan) is cheaper and widely used: its controlled attenuation parameter (CAP), reported from 100 to 400 dB/m, grades steatosis, with cut-offs of 248, 268 and 280 dB/m for grades S1, S2 and S3 derived from an individual patient data meta-analysis [11].

CAP is, however, an imperfect measure of liver fat. A review of the technique reports that agreement between two operators examining the same patient has a concordance correlation of 0.82, falling to 0.65 in patients with BMI of 30 $\mathrm{kg/m^2}$ or more and to 0.44 when CAP is below 240 dB/m [12]. The same review notes that CAP accuracy falls when the interquartile range (IQR) of the ten acquisitions exceeds 40 dB/m, which is why that limit is used as a quality criterion. In adults with NAFLD, CAP correlated with MRI-PDFF at $r = 0.66$, falling to $r = 0.37$ for changes over time [61]. Part of every CAP reading is therefore measurement variation, and the variation is largest in exactly the people who are hardest to examine. This limits how closely any model can predict CAP (Section 2.3.3).

Simple clinical scores offer a cheaper alternative. The Fatty Liver Index (FLI) combines BMI, waist circumference, triglycerides and GGT [47]; the Hepatic Steatosis Index (HSI) combines the ALT/AST ratio, BMI, sex and diabetes [48]; and the NAFLD liver fat score combines metabolic syndrome, type 2 diabetes, fasting insulin, AST and the AST/ALT ratio, reaching an AUC of 0.86–0.87 against spectroscopy, with genetic information adding less than 1% [62]. These scores are the natural baselines for any new steatosis model.

2.3 Machine-Learning Prediction of Steatosis and Liver Fat

2.3.1 Classification of steatosis in NHANES

The US National Health and Nutrition Examination Survey (NHANES) added FibroScan to its 2017–March 2020 cycle, and several studies have since trained classifiers to detect CAP-defined steatosis from routine data (Table 2.1). Gradient-boosted trees, usually XGBoost [38], perform best or near best in most of them, and the strongest predictors are consistently body-size measures (waist circumference and BMI), followed by liver enzymes, glucose and insulin markers, and triglycerides. Model explanations, typically SHAP values [40], are widely reported.

Du et al. used 6,814 participants and found logistic regression, XGBoost and LightGBM comparable (AUC 0.815–0.842), so that the added complexity of boosting bought little [49]. Liu et al. compared models on an automated machine-learning platform in 4,177 participants; XGBoost reached an AUC of 0.859, and the authors imputed missing values by multiple imputation and reported calibration curves and decision-curve analysis [50]. Zhang et al. reported an AUC of 0.874 on 2,460 complete cases [51]. Boullion et al. identified biomarkers of steatosis and stiffness; propensity matching on age and sex required complete data and reduced the steatosis sample to 2,714, so their models were trained on the larger unmatched data, where random forests reached 79.6% accuracy [52]. Zhu et al. trained a three-grade model on 215,145 non-obese Chinese adults, reaching a macro-average AUC of 0.941 internally and 0.874 in 726 non-Hispanic Asian NHANES participants with complete data [63]; this is one of the few studies with external validation, and it shows the usual fall in performance when a model moves to a new population.

Three limitations recur. Most studies use only the 2017–2020 cycle, and most discard participants with any missing value. Calibration is reported only occasionally [50], although a risk model whose

Table 2.1: Machine-learning classifiers of CAP-defined steatosis using NHANES.

Study	Population	Outcome	Best model	Performance
Du et al. (2026) [49]	6,814 adults, 2017–2020	MASLD, CAP $\geq$ 280	LR, XGBoost, LightGBM	AUC 0.815–0.842
Liu et al. (2024) [50]	4,177, fasting subsample; imputed	CAP $\geq$ 302	XGBoost (AutoML)	AUC 0.859; calibration curve
Zhang et al. (2025) [51]	2,460 complete cases	CAP-defined MASLD	XGBoost	AUC 0.874
Boullion et al. (2025) [52]	Unmatched sample (matched subset 2,714)	CAP $\geq$ 238	Random forest	Accuracy 79.6%
Zhu et al. (2026) [63]	215,145 non-obese Chinese adults; 726 NHANES Asians	CAP grades (S0, S1, S2–S3)	XGBoost	Macro AUC 0.941 internal; 0.874 external

output will be shown to patients must produce probabilities that mean what they say; isotonic regression is a standard way to obtain them from tree ensembles [39]. And all are single snapshots, used to classify a participant once rather than to track or support them over time.

2.3.2 Prediction of liver fat as a continuous value

Fewer studies predict the amount of liver fat, and their results are more sobering (Table 2.2). In 28,796 UK Biobank adults, the Fatty Liver Index explained 28% of the variance in MRI-PDFF (33–37% across cross-validation folds); an extended model with twelve variables (FLI+) explained 54% on average across four cross-validation folds (0.520–0.557) [64]. A UK Biobank preprint reported R^2 of 0.37 from anthropometric and blood measures in 40,533 participants with MRI-PDFF and 0.34 from plasma metabolites in 9,723, against 0.63–0.64 from DXA body-composition images analysed by a convolutional network [65]. Haas et al., also in UK Biobank, found that BMI correlated with MRI liver fat at $r = 0.42$ and that adding 24 clinical factors and biomarkers raised the correlation only to $r = 0.58$ ($R^2 \approx 0.34$), concluding that liver fat cannot be reliably estimated from clinical factors [66]. In Finnish adults measured by spectroscopy, 15–25% of the variance could be alone explained by liver enzymes; an equation including a specialist marker (phosphorylated IGFBP-1) reached an adjusted R^2 of 0.49 in validation and an AUROC of 0.84 for NAFLD, against 0.72 for FLI and 0.62 for HSI [67]. Higher values come from small, selected samples: in 272 patients of a Chinese obesity clinic, a four-variable equation predicted CAP with $R^2 = 0.76$ in derivation and 0.65 in validation [53], and a four-variable index correlated with MRI-PDFF at $\rho = 0.74$ in 149 overweight or obese adults, validated only by bootstrap resampling [68]. No study reporting R^2 for CAP in NHANES was found.

2.3.3 Why prediction from routine data is bounded

Two findings explain why continuous liver fat is hard to predict from routine measurements. First, liver fat is strongly heritable: in a prospective study of monozygotic and dizygotic twins using MRI-PDFF, the heritability of steatosis was 0.52 after adjustment for age, sex and ethnicity [69], and in

Table 2.2: Prediction of liver fat as a continuous value.

Study	Population	Reference	Inputs	R^2
UK Biobank preprint (2024) [65]	40,533; 9,723; about 46,000 UK adults	MRI-PDFF	Body and blood measures; metabolites; DXA images	0.37; 0.34; 0.63–0.64
Asaturyan et al. (2022) [64]	28,796 UK adults	MRI-PDFF	FLI; FLI with 12 variables (FLI+)	0.28–0.36; 0.54 (4-fold mean)
Haas et al. (2021) [66]	UK adults, 1,214 test	MRI	BMI; BMI with 24 clinical factors	r 0.42; r 0.58
Petäjä et al. (2016) [67]	378 Finnish adults	^{1}H-MRS	Enzymes only; equation with phosphorylated IGFBP-1	0.15–0.25; 0.49 (adjusted)
Liu et al. (2022) [53]	272, obesity clinic	CAP	BMI, neck circumference, TG, C-peptide	0.76 / 0.65
Chen et al. (2023) [68]	149, overweight or obese	MRI-PDFF	ALT, AST, insulin, 1-h glucose	$\rho = 0.74$
Kotronen et al. (2009) [62]	470 Finnish adults	^{1}H-MRS	Metabolic and liver markers	Not reported

UK Biobank measured genetic variants together explained up to 30% of the variance in liver fat [66]. Variants such as those in *PNPLA3* change liver fat at the same body size, and routine clinical data contain none of this information. Second, the reference measurement itself is noisy (Section 2.2): a model can explain no more of the variation in a single CAP reading than the reading shares with the patient's true liver state. Together, these findings imply that a model built on routine data should reach a moderate R^2 for liver fat, between about 0.3 and 0.55 in the large studies above, and good discrimination of steatosis, which is the pattern seen in Tables 2.1 and 2.2. For a digital twin, the implication is that a calibrated probability of steatosis is a more defensible state variable than a single predicted CAP value.

2.4 Dietary Assessment and Personalised Nutrition

Personalising dietary advice requires knowing what a person usually eats. NHANES collects one or two 24-hour dietary recalls per participant. A single recall describes the average intake of a group well, but an individual's usual intake poorly, because intake varies from day to day. Statistical methods developed at the US National Cancer Institute estimate usual intake from two or more recalls by separating within-person from between-person variation [44]. For foods that are not eaten every day, the NCI method uses a two-part model that separates the probability of eating a food on a given day from the amount eaten on days it is eaten, allows the two parts to be correlated, and can add covariates such as age, sex or a food frequency questionnaire; in simulation it estimated the distribution of usual intake substantially better than earlier methods [45]. Self-reported intake is also biased: in the OPEN study, comparison with recovery biomarkers showed substantial under-reporting, particularly of energy [43]. Diet quality can be summarised with the Healthy Eating Index-2020, a 0–100 score of 13 components aligned with the 2020–2025 Dietary Guidelines for

Americans; its components are unchanged from HEI-2015 for ages 2 and over, and a separate index was introduced for toddlers [70]. Computing it requires food-group data from the USDA Food Patterns Equivalents Database.

Personalisation is motivated by individual variation in response. Zeevi et al. showed in 800 people that post-meal glucose responses to identical meals differ widely and can be predicted from personal data, and that diets personalised on these predictions lowered post-meal glucose [19]. For a liver twin, these findings suggest two requirements: rather than single-day intake, dietary inputs should represent the usual intake, and the system should become more personal as more of the patient's own data arrive.

2.5 Digital Twins in Health and Nutrition

2.5.1 Definitions and requirements

The US National Academies define a digital twin as a set of virtual information constructs that mimics the structure, context and behaviour of a physical system, is dynamically updated with data from it, has a predictive capability and informs decisions; the bidirectional flow between the virtual and physical system is central, and credibility rests on verification, validation and uncertainty quantification (VVUQ) [14]. A scoping review of health twins frames them as a physical entity, its virtual representation and the connection between them, and found that the term is used for systems ranging from static models to fully coupled ones [16]. A perspective on medical twins describes them as most advanced in cardiology and oncology, and names data integration, validation and privacy as the main obstacles [15].

Two architectural proposals are relevant to a modular twin. Nitschke et al. propose a general patient twin built on a bipartite knowledge graph that links patient attributes to existing predictive models; fusion models based on ensemble learning combine the models' outputs, and a propagation scheme passes information through the graph each time new patient data arrive. They describe the design as modular, evolving, informed, interpretable and explainable [71]. Nye structures patient data as a knowledge graph built on medical ontologies such as SNOMED CT and proposes closed-form continuous-time ("liquid") neural networks to model the patient's evolving state [72]. Both are conceptual designs without empirical evaluation, but they share a principle that HepatoTwin adopts: separate modules, each with a defined role, connected by an explicit patient state.

2.5.2 Metabolic and nutrition twins

In nutrition, Gkouskou et al. extended the idea of an "actual digital twin", in which a patient is matched to many similar patients whose outcomes are known, into a "virtual digital twin": a personal model combining genetic, longitudinal metabolomic, immune, microbial and behavioural data, collected through connected devices and managed by machine learning, to simulate dietary interventions for that person [20]. Metabolic twins have also been outlined as tools that track

a person's metabolic flexibility over time [17]. A systematic review of twins in precision health identified 12 studies and found improvements in most of the reported outcomes, while noting that the evidence is still small and heterogeneous [18].

The most developed clinical example is the Twin Health platform. Its method paper describes a machine-learning model (CatBoost) that predicts each person's post-meal glucose response from continuous glucose monitoring, food logs, activity and sleep, and a mobile application that adjusts dietary recommendations from those predictions and keeps refining them as data arrive [22]. In a one-year randomised trial in 319 people with type 2 diabetes, nutrition guided by this twin improved liver-fat scores and MRI-PDFF compared with standard care [21]. A second example, iDT-diet, modelled how diet quality affects biomarkers over time in 402 participants measured at baseline and after one, three and six years. It combined a random forest with Choquet-integral LASSO feature selection, which had lower error than XGBoost, LightGBM and CatBoost for several targets, with generated natural-language descriptions and three-dimensional avatars of the predicted body state [23]. In MASLD specifically, the European ARTEMIS project is building virtual twins from retrospective data on several thousand patients [24].

These systems recommend individual foods from predicted glucose responses/forecast biomarker trajectories. The Twin Health platform depends on continuous glucose monitoring and proprietary data, and iDT-diet forecasts biomarkers without recommending food. None evaluates complete, existing recipes against evidence-graded dietary priorities, and none combines health with cuisine and practicality when making a recommendation.

2.5.3 From prediction to "what if"

A twin that compares interventions must answer counterfactual questions, which can not alone be licensed by predictive accuracy. Freeman et al. name two traps. The fidelity trap is the belief that a model which is accurate enough to reproduce patient trajectories can therefore say what would happen under a different treatment; prediction and counterfactual estimation are distinct tasks with different justifications. The feedback trap arises when a twin learns from data that its own recommendations helped to produce, which breaks the assumptions behind its causal estimates. They argue that health twins should be causally valid, modular and evolving systems, with modularity isolating the parts that need interventional justification [73]. According to Laudy, a digital twin should represent how an individual might respond under treatment and control scenarios. To assess its reliability, the framework defines three dimensions of fidelity—marginal, joint, and structural—and outlines five levels of validation through which model predictions can be examined using observable data [74]. However, Laudy acknowledges that the relationship between an individual's potential outcomes under the two conditions cannot be directly verified [74].

For HepatoTwin, these arguments have a direct consequence. A cross-sectional model trained on observational data, such as a steatosis classifier built on NHANES, describes the current state well but cannot by itself predict the effect of changing a diet. Intervention effects must instead come from interventional evidence, held in a separate module with their strength and uncertainty made

explicit, and the twin's updating must be evaluated in a setting where the true effect is known (Section 2.7).

2.6 Computational Gastronomy and Food Recommendation

2.6.1 Recipe and food resources

Computational gastronomy treats food and cooking as structured data, studying recipes through their flavour, nutrition, health associations and sustainability [25]. RecipeDB contains 118,171 recipes from six continents, 26 geocultural regions and 74 countries, cooked with 268 processes from over 20,000 ingredients, with cuisine labels, utensils and nutritional profiles, and ingredients linked to their flavour molecules [26]. FlavorDB catalogues 25,595 flavour molecules, 2,254 of which are linked to 936 natural ingredients in 34 categories [27]. Literature mining of 5,769 abstracts produced 8,957 associations between 152 culinary spices and 848 diseases; for 150 of the 152 spices the beneficial associations outnumbered the adverse ones, and the authors used this balance to make disease-specific spice recommendations [28]. These resources make it possible to reason about recipes at scale, but they are built for exploration and analysis rather than for deciding which recipes suit a particular patient.

2.6.2 Food recommender systems

Food recommendation faces challenges beyond those of other domains: it must balance taste, health, culture and practicality. In a survey of the field, Trattner and Elsweiler showed that re-weighting standard recommender outputs by WHO or UK Food Standards Agency health scores can trade accuracy for healthiness, but the healthiest lists came with very poor accuracy, and they named health-aware recommendation an open problem [30].

Knowledge graphs are a common foundation. FoodKG links over a million recipes, their ingredients and USDA nutrient data in 67 million triples, and supports queries such as which recipe can be made from the ingredients at hand while avoiding allergens [31]. Chen et al. framed personalised recommendation as constrained question answering over FoodKG, treating the user's query, dietary preferences and health guidelines as constraints, and improved on non-personalised baselines by 59.7 percentage points on average across metrics [32]. Graph neural networks have been used to move users gradually towards healthier recipes while respecting their preferences [33]. In all of these systems, "healthy" is defined by generic nutritional guidelines rather than by a patient's clinical state, and the evidence behind each recommendation is not graded.

2.6.3 Recipe modification and adaptation

A related line of work modifies recipes by substituting ingredients. Shirai et al. combined explicit knowledge from a food knowledge graph with word embeddings into a substitutability heuristic (DIISH), filtered the candidates by nutritional and food-category constraints, and evaluated it

against substitutions taken from cooking guides and recipe reviews [34]. Loesch et al. learned knowledge-graph embeddings (RDF2Vec) over a food graph and evaluated them against 1,847 known substitution pairs, retrieving about 40% of the true substitutes in the top ten candidates [75]. Rita et al. fine-tuned large language models to choose substitutions that increase phytochemical content, improving Hit@1 from 34.5% to 38.0% on the GISMo benchmark and producing 1,639 enriched recipes, while cautioning that the health claims rest on preclinical evidence [36]. A survey of the area concludes that a good substitute must match flavour, culinary function and health suitability together, and recommends knowledge graphs and neuro-symbolic reasoning for doing so [35]. Cultural adaptation goes further: translating recipes between Chinese and English cuisines requires changing ingredients, units, tools and techniques, and GPT-4 still fell short of human experts when adapting English recipes into Chinese [37].

Every modification creates a new recipe whose taste, feasibility and nutrition must be predicted, and none of these studies tested whether people liked or cooked the result. Evaluating existing recipes, the approach taken in this thesis, avoids that uncertainty: every recommended recipe has already been written, cooked and nutritionally described.

2.6.4 Menu planning, diversity and additional criteria

Menu planning composes meals that satisfy nutritional and cost targets; Sahin et al. released EvoMeal, an open-source tool comparing four multi-objective evolutionary algorithms, and found that two newer algorithms (AGE-MOEA and SMS-EMOA) outperformed NSGA-II and NSGA-III [29]. Recommendation lists also benefit from diversity: maximal marginal relevance re-ranks items by a linear combination of relevance and dissimilarity to the items already chosen, also reducing redundancy [76]. Two further criteria are relevant to MASLD guidance. The degree of food processing can be estimated with FPro, a machine-learning score that predicts processing from nutrient profiles; it classified over 73% of the US food supply as ultra-processed, was associated in NHANES with metabolic syndrome and diabetes, and showed that swapping foods for less-processed alternatives could reduce these associations [77]. Glycaemic load is available from the international glycaemic index tables [78].

2.7 Evaluating Digital Twins Without Longitudinal Data

A twin's defining features, updating with new data and projecting the future, can only be fully validated against repeated measurements of the same people. When such data are unavailable, simulation offers a recognised alternative. In-silico clinical trials test interventions on computer-simulated patients before, or alongside, real trials [54]. The best-known example is the UVA/Padova type 1 diabetes simulator, a population of virtual patients built from physiological models, which the US Food and Drug Administration accepted in 2008 as a substitute for animal trials in the preclinical testing of closed-loop insulin control [55], [79]. Its authors stress that such testing is a prerequisite to clinical trials, not a substitute for them [55]. Synthea generates synthetic patient

records for software testing at scale [56].

Two designs are particularly relevant. In plasmode simulation, real data provide the covariates and the outcome is simulated from a known model, so the data keep their real structure while the true effect is known and a method's ability to recover it can be measured [57]. In data assimilation, observing system simulation experiments generate a "true" state with one model, give a forecasting system only noisy observations of it, and measure how well the system recovers the truth; the field calls these twin experiments [46].

Simulated follow-up is credible only if its ingredients come from evidence. For weight change driven by diet, Hall et al. built and validated a dynamic model that links a sustained change in energy intake to body weight over time. In their model, cutting intake by about 10 kcal per day for good lowers weight by about 0.45 kg in the long run; half of that change comes within about a year and 95% within about three years. The familiar static rule of 3,500 kcal per pound overstates long-term weight loss by about 100%, and the model predicts average rather than individual responses [80].

Large lifestyle trials report the weight trajectories that real participants achieve. In Look AHEAD, intensive lifestyle intervention produced a mean weight loss of 8.6% at one year against 0.6% with usual care; 68.0% and 37.7% of participants lost at least 5% and 10%, and at eight years the mean losses were 4.7% and 2.1% [81]. The Diabetes Prevention Program reported a similar pattern of loss and partial regain [82], and a Look AHEAD analysis showed that losses of 5–10% increased the odds of clinically meaningful improvements in glycaemia, blood pressure and lipids [83]. The response of the liver itself is given by the weight-loss studies of Section 2.1 [5], [59], [60]. Realistic observations also require the normal biological variation of each laboratory test [84] and the known under-reporting of food intake [43].

These sources make it possible to evaluate a twin's updating on virtual patients whose follow-up is grounded in published human data. Such an evaluation verifies that the twin behaves correctly when the truth is known; it does not establish its accuracy for real patients, which still requires longitudinal data.

2.8 Summary and Research Gap

Table 2.3 summarises what each strand of the literature offers and what it leaves open. Clinical studies have established links between dietary changes, weight management, and liver health. However, their findings are generally reported in terms of energy intake, nutrient composition, and dietary patterns rather than individual recipes. Machine-learning models have shown promise in identifying hepatic steatosis using routinely collected clinical data, yet most are designed for one-time classification, with limited calibration and constraints on continuous prediction arising from genetic variability and measurement uncertainty.

Existing digital twins for nutrition support personalised food choices or predict changes in biomarkers, but they do not assess complete recipes against priorities ranked by the strength of clinical evidence. Moreover, causal inference research shows that a model's ability to predict outcomes

Table 2.3: Related research areas, their remaining limitations, and the contribution of HepatoTwin.

Strand	What it provides	What it leaves open	HepatoTwin
Clinical evidence (Section 2.1)	Graded effects of weight loss, diet patterns and nutrients	Not expressed for recipes; evidence strength varies	Evidence engine with explicit strength labels, mapped to recipe criteria
Steatosis models (Section 2.3)	Good discrimination from routine data	Snapshots; rarely calibrated; bounded R^2	Calibrated probabilities as the twin's state, within a defined scope
Dietary assessment (Section 2.4)	Recall data; usual-intake methods	Single-day recalls used directly	Recall-adjusted priorities; usual intake updated from food logs
Digital twins (Section 2.5)	Personal state, glucose response prediction, trajectories	No recipe-level, evidence-graded recommendation; fidelity and feedback traps	Modular twin linking state, evidence and recipes; effects from interventional evidence
Food recommendation (Section 2.6)	Recipe data, preferences, generic healthiness, substitution	No clinical state; ungraded evidence; untested modified recipes	Evidence-gated ranking of existing recipes by health, cuisine and feasibility
In-silico evaluation (Section 2.7)	Virtual patients, plasmode and twin experiments	Not yet applied to liver twins	In-silico evaluation of the twin's updating on virtual NHANES patients

accurately does not, by itself, demonstrate that its recommended interventions will lead to the expected results. Food recommendation and ingredient substitution systems can account for user preferences and general measures of nutritional quality, but typically lack integration with individual clinical conditions and the certainty of supporting evidence. In addition, the performance of liver-specific digital twins has not yet been adequately assessed in terms of how their predictions and recommendations evolve as new patient data become available.

This thesis sits where these strands meet. It builds a liver twin for each patient and combines the twin's calibrated state with graded intervention evidence and the patient's own diet to rank existing recipes, so that every ranking can be traced back to its evidence. Because real follow-up data do not yet exist, the twin's updating is tested on simulated patients.

Chapter 3

Experiments on Predicting Liver Fat

This chapter records the experiments run to predict the controlled attenuation parameter (CAP; NHANES variable LUXCAPM, dB/m), the FibroScan measure of liver fat that the patient-state model of HepatoTwin estimates from routine data. Fifty-one experiments, run between 5 September and 22 October 2026, varied the learning algorithm, the inputs, the training weights, the treatment of the target, the training data and the evaluated population. Section 3.1 defines how results were measured; Section 3.2 lists every experiment and its result; Section 3.3 describes the model that was selected and its performance; Section 3.4 explains why R^2 stops rising; and Section 3.5 compares the result with published models. Section 3.6 summarises the chapter.

3.1 Evaluation Protocols

The cohort comprised NHANES participants with a complete elastography examination (LUAXSTAT $= 1$) and an observed CAP, aged 12 to 80 years. R^2 is the share of the variation in measured CAP that the predictions explain on participants the model was not trained on. Three protocols were used as the work progressed (Table 3.1). Because each protocol scores a different set of participants, R^2 values from different protocols are not directly comparable, and every result below is labelled with its protocol.

Outcomes are labelled *Adopted* (carried into the model), *Small gain* (below about 0.01, not or not fully adopted), *No gain*, *Worse* and *Diagnostic* (an analysis explaining the model's behaviour rather than a candidate model).

3.2 Experiments

The experiments fall into six phases. Each phase was prompted by the findings of the one before it: once changes to the model stopped helping, the work turned to the target variable, then to new data, and finally to the population in which the model is used.

Table 3.1: Evaluation protocols.

Code	Design	Used for
P1	9,021 participants (2017–March 2020) split once 80/20 (seed 42) into 7,216 development and 1,805 test participants; model choice by five-fold cross-validation on the development set (P1-CV), test results as P1-test.	September and early October. The test set was examined repeatedly, so P1-test is an internal comparison.
P2	All 9,021 participants, five-fold cross-validation (seed 42); R^2 on the pooled out-of-fold predictions. Baseline 0.5135.	Model families and new data (October).
P3	15,300 participants from both FibroScan cycles (2017–2020 and 2021–2023). A 20% test set (3,060, stratified by cycle, seed 42) set aside first; model choice by five-fold cross-validation on the remaining 12,240 (P3-CV); test set scored once (P3-test).	Combined model and final selection.

Table 3.2: Phase 1 and 2 experiments: baselines, tuning and inputs.

ID	Experiment	Result	Outcome
E1	Baseline XGBoost, 14 inputs	P1-test R^2 0.491; RMSE 44.5 dB/m	Adopted
E2	XGBoost against linear regression and random forest	P1-test 0.491; 0.488; 0.486	Small gain
E3	Leakage-free preprocessing and a 40-setting search	Corrected baseline 0.485 → 0.506 (P1-test)	Adopted
E4	14, 19 and 28 inputs (extra blood tests) and 24 model variants	P1-CV 0.489 / 0.496 / 0.501; blend 0.502	Small gain
E5	Reproduction of a supplied model reporting 0.535	On its own split: supplied 0.535, ours 0.541; on our split: 0.511 and 0.514	Diagnostic
E6	36 raw inputs plus 10 engineered features	P1-CV 0.506; P1-test 0.515	Small gain

3.2.1 Phases 1 and 2: baselines, tuning and inputs

The first models used 14 inputs: age, sex, race and ethnicity, BMI, waist circumference, ALT, AST, HbA1c, total and HDL cholesterol, and day-one dietary energy, protein, carbohydrate and fat (Table 3.2).

A linear model came close to the tree ensemble (E2), a first sign that the signal is mostly additive and driven by body size. The first clear gain came from fitting imputation inside each fold and tuning the trees (E3). Shallow trees, a low learning rate and row subsampling reduce overfitting when the target is noisy. Rather than coming from a better model, experiment E5 showed that a supplied model's higher R^2 came from an easier test split, and that a single 20% split moves R^2 by about ± 0.03.

3.2.2 Phase 3: features, weighting and learners

Phase 3 tested whether reshaping the inputs or the training weights could help (Table 3.3). None changed R^2 by more than 0.003.

Table 3.3: Phase 3 experiments: feature ablations and weighting (P1-CV).

ID	Experiment	Result	Outcome
E7	Engineered-feature ablation	0.5035 / 0.5048 / 0.5032 / 0.5036	Small gain
E8	Single ratios (WHtR, AST/ALT, TG/HDL, TC/HDL)	0.5022–0.5038 against 0.5035	No gain
E9	LDL cholesterol	0.5036 → 0.5020 / 0.5030	No gain
E10	Recursive feature elimination, 46 to 10 inputs	46: 0.504; 25: 0.508; 15: 0.501; 10: 0.485	No gain
E11	Histogram gradient boosting	0.486 against XGBoost 0.505	Worse
E12	Up-weighting extreme CAP values	0.505 → 0.500 / 0.494 / 0.488	Worse

Table 3.4: Phase 4 diagnostics of the target.

ID	Experiment	Result	Outcome
E13	Residuals by CAP range (P2)	Slope of prediction on actual 0.51; bias +44 dB/m below 200 and −67 dB/m above 390	Diagnostic
E14	CAP ceiling audit	129 of 9,021 (1.4%) exactly 400 dB/m	Diagnostic
E15	Refit without CAP = 400	R^2 0.506 → 0.488; RMSE 44.3 → 43.7 dB/m	No gain
E16	Performance by probe	Within-probe R^2: M 0.419, XL 0.278	Diagnostic

About 20 to 25 inputs carry nearly all the signal, led by waist circumference, waist-to-height ratio and BMI (E10). Derived scores such as the Fatty Liver Index re-express inputs that the trees can already combine (E7–E8).

3.2.3 Phase 4: the target and other model families

Phase 4 first examined the target itself (Table 3.4) and then tried other model families and target treatments (Table 3.5).

Predictions are pulled towards the mean of about 255 dB/m (E13): the inputs cannot separate very lean or very fatty livers from average ones. The device truncates CAP at 400 dB/m (E14), but too few readings are affected to limit R^2, and removing them lowers R^2 because the outcome variance shrinks (E15).

Every model family reached the same ceiling (E19, E24–E26). The binary classification of steatosis (E27) performed well and became the second output of the model; the four-class task failed for the middle grades, which span only 20 and 12 dB/m, less than the model error.

3.2.4 Phase 5: additional data

Phase 4 indicated that the limit lay in the information available, not in the model, so Phase 5 added NHANES data not used before (Table 3.6). Elastography fields and self-reported liver conditions were never used as inputs, because they would leak the target.

The largest single gain in the record, +0.010, came from the extra NHANES variables (E29), mainly

Table 3.5: Phase 4 experiments: model families and target treatments.

ID	Experiment	Result	Outcome
E17	Probe type (M or XL) as an input	0.5062 → 0.5112	Small gain[a]
E18	XL examinations weighted 1.5× and 2×	0.5112 → 0.5080 / 0.5018	Worse
E19	CatBoost (P2)	0.5093 against XGBoost 0.5135	No gain
E20	Steatosis-class probabilities fed to the regressor	0.5116 / 0.5129	No gain
E21	Linear recalibration of predictions	0.5127	No gain
E22	Quantile (median) regression	0.503 against 0.506; 80% interval covers 74.7%	No gain
E23	Fold-local Lasso and RFE selection	0.499; 0.504	No gain
E24	Extra Trees, random forest, ridge	0.500; 0.498; 0.482	No gain
E25	Neural network (128, 64)	0.453	Worse
E26	Stacking with a ridge meta-learner	0.506 → 0.509	Small gain
E27	Steatosis classification	Binary AUC 0.862; four-class macro F1 0.439	Adopted
E28	HbA1c ablation	Not completed	–

[a]Not used: probe type is recorded during the FibroScan examination itself.

insulin resistance and diabetes and weight history. The learning curve was flat (E30), so a second cycle of 6,279 participants added only 0.004 (E31). Even visceral fat measured directly by DXA, the strongest known correlate of liver fat, added only 0.007 (E32), and inputs taken from the FibroScan examination itself reached only 0.53 (E34). The combined model of E35 pooled both cycles and down-weighted less reliable examinations; it was adopted.

3.2.5 Phase 6: target population and lean participants

The cohort filters of E36 showed that R^2 is highest among participants aged 12 to 59 with a reliable examination (CAP interquartile range below 40 dB/m). In this group the error itself falls (RMSE 43.6 → 41.6 dB/m) while the spread of CAP does not shrink, so the higher R^2 reflects better predictions, not an easier outcome. On the supervisor's advice this group was adopted as the model's scope, and Phase 6 tried to reach 0.60 within it (Table 3.7).

All variants scored between 0.584 and 0.597, a spread smaller than the variation between cross-validation folds (0.546 to 0.617). Training on the scope alone was worse than training on everyone (E40): participants outside the scope still carry information about how the inputs relate to CAP. Discrimination in lean participants (E48–E51) fell in every BMI band, and the Fatty Liver Index fell by a similar amount, because within a narrow BMI band the strongest predictor, body size, hardly varies.

3.3 Final Model and Results

The selected model is an XGBoost regressor [38] with 3,000 trees, depth 5, learning rate 0.005, minimum child weight 15 and column subsampling 0.4, trained on the 12,240 training participants of both FibroScan cycles. It uses 81 routine inputs (body measurements, standard blood tests,

Table 3.6: Phase 5 experiments: additional data.

ID	Experiment	Result	Outcome
E29	About 80 more NHANES variables (insulin, CRP, blood count, blood pressure, questionnaires, prescriptions)	0.5135 → 0.5239 (128 inputs, P2)	Small gain
E30	Learning curve (25–100% of training data)	0.488; 0.501; 0.511; 0.512	Diagnostic
E31	2021–2023 FibroScan cycle (6,279 participants)	0.5135 → 0.5170	Small gain
E32	DXA visceral and subcutaneous fat (3,307)	0.5094 → 0.5168	Small gain
E33	Measurement noise within an examination	About 3.7% of CAP variance	Diagnostic
E34	Upper bound with FibroScan-derived inputs	0.518 → 0.524 (probe), 0.529 (stiffness)	Diagnostic
E35	Combined model: both cycles, 81 inputs, reliability weights (P3)	P3-test 0.480 → 0.493; P3-CV 0.520	Adopted
E36	Cohort filters	All 0.518; IQR < 40 and ages 12–59: 0.584	Diagnostic
E37	Performance by age group	12–17: 0.43; 18–39: 0.54; 40–59: 0.49; 60+: 0.41	Diagnostic
E38	Improving the 60+ group	0.405 → 0.394–0.408	No gain
E39	Calibrated classifiers, CAP ≥ 248 and ≥ 280	AUC 0.856 and 0.868; calibration error 0.025 → 0.018	Adopted

blood pressure and a short questionnaire) and six derived scores: waist-to-hip ratio, HOMA-IR, the triglyceride-glucose index, mean systolic and diastolic blood pressure, and one-year weight change. Examinations with a CAP interquartile range of 40 dB/m or more are down-weighted (weight 40/IQR, at least 0.25) rather than dropped. Two isotonic-calibrated classifiers estimate the probability that CAP is at least 248 and at least 280 dB/m.

The tree settings were chosen by scoring each of the 15 Phase 6 candidates on the in-scope members of the held-out test set; the deepest setting scored highest (0.5622 against 0.5582 for the reference). It also has the highest full-population cross-validation R^2 of the 12 settings compared, and repeated cross-validation, which played no part in the choice, confirms the gain (0.593 against 0.590). Because the test set informed the choice, the test figures are slightly optimistic.

Within the scope the held-out test R^2 is 0.562 (95% bootstrap interval 0.517–0.601; Table 3.8), and 0.593 in five-fold cross-validation over all 15,300 participants repeated with three seeds. The value is lower within each age band than in the whole scope because CAP varies less within a band. Further checks did not change the picture (Table 3.9): the result holds when weighted to the US population, and a model trained on 2017–2020 alone predicted the 2,415 in-scope participants of 2021–2023 with R^2 0.578.

On the 1,100 in-scope test participants with every input to both scores, the CAP ≥ 248 classifier reached AUC 0.881 against 0.861 for the Fatty Liver Index [47] and 0.836 for the Hepatic Steatosis Index [48]; the paired differences (+0.020 and +0.046) both exclude zero.

Table 3.7: Phase 6 experiments within the scope (P3-CV; reference 0.5945).

ID	Experiment	Result	Outcome
E40	Training only on the scope; scope weighted 3×; ages 12–59 only	0.5845; 0.5921; 0.5883	Worse
E41	Habitual diet from the 24-hour recall	0.5930	No gain
E42	Medical history and prescription count	0.5949	No gain
E43	Fasting subsample only	0.597 against 0.592 (test: 0.550 against 0.560)	No gain
E44	Median imputation with indicators; MICE	0.5961; 0.5938	No gain
E45	Pseudo-Huber loss; absolute-error loss	0.5940; 0.5893	No gain
E46	Five further tree settings	0.5915–0.5960	Adopted[b]
E47	Ensembles of 2–5 variants	Best 0.5969	Small gain
E48	Lean-only classifier (BMI < 28)	Lean AUC 0.804 against 0.809	Worse
E49–E51	Lean weighting, non-obese training, blend	Lean AUC 0.807–0.808	No gain

[b]The deepest setting had the highest test R^2 of all candidates (Section 3.3).

Table 3.8: Selected model on the held-out test set (P3-test). The reference uses the 36 core inputs on the same participants.

Test group	n	R^2	Reference R^2	RMSE (dB/m)
Scope: ages 12–59, reliable reference	1,236	0.562	0.553	41.4
Ages 12–17	209	0.498	0.482	36.2
Ages 18–39	512	0.518	0.514	42.7
Ages 40–59	515	0.522	0.506	42.1
Outside the scope: ages 12–59, all examinations	2,040	0.515	0.511	–
Outside the scope: all participants	3,060	0.496	0.487	43.5

3.4 Why R^2 Plateaus

Nine independent lines of evidence show that the remaining variation in CAP cannot be explained by routine clinical measurements:

1. Every algorithm tested (linear regression, random forest, Extra Trees, histogram boosting, XGBoost, CatBoost, a neural network and stacking) reaches R^2 between 0.45 and 0.52 on the same inputs (E2, E11, E19, E24–E26).
2. Feature engineering, selection and ablation change R^2 by at most 0.003 (E7–E10, E23).
3. The learning curve is flat: 6,279 additional participants add 0.004 (E30, E31; Figure 3.1).
4. Direct DXA measurement of visceral fat adds 0.007 (E32).
5. About 80 additional NHANES variables add 0.010 (E29).
6. Even inputs taken from the FibroScan examination itself reach only 0.53 (E34).
7. Noise within one examination is about 4% of the CAP variance (E33), so the unexplained variance reflects real differences between people with similar measured profiles.

Table 3.9: Selected model within the scope, with 95% bootstrap intervals.

Measure	Held-out test	Survey-weighted	Temporal (2021–2023)
CAP R^2	0.562 (0.517–0.601)	0.554 (0.493–0.609)	0.578 (0.549–0.603)
CAP RMSE (dB/m)	41.4 (39.3–43.6)	42.6 (39.5–45.7)	42.9 (41.4–44.5)
AUC, CAP $\geq$ 248	0.881 (0.863–0.899)	0.877 (0.851–0.901)	0.884 (0.870–0.896)
AUC, CAP $\geq$ 280	0.879 (0.860–0.898)	0.878 (0.849–0.905)	0.890 (0.878–0.904)

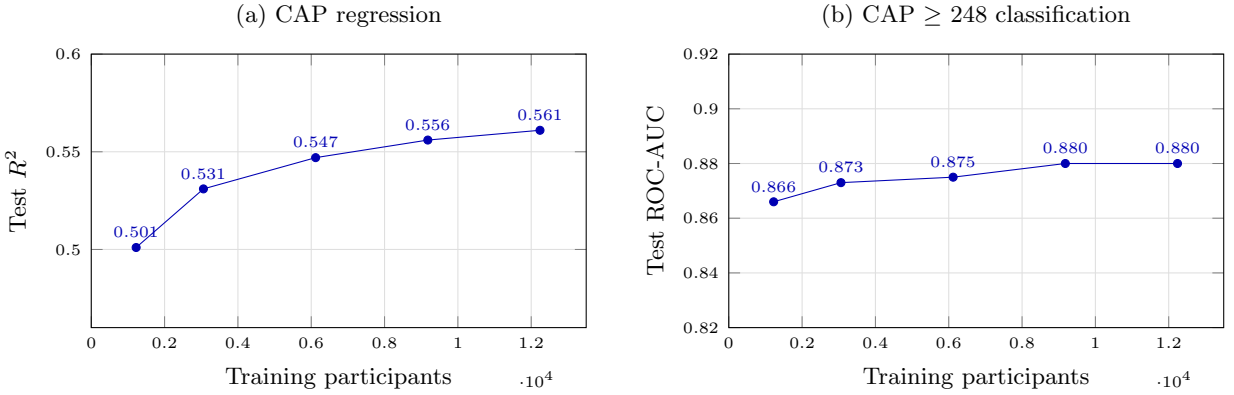


Figure 3.1: Learning curve of the selected settings: held-out test R^2 and AUC within the scope ($n = 1{,}236$) against the number of training participants. R^2 rises from 0.501 with 1,224 participants to 0.556 with 9,180 and only to 0.561 with all 12,240.

8. Within the scope, every change to inputs, imputation, loss, tuning, training population and ensembling scored between 0.584 and 0.597, a spread smaller than the variation between folds (E40–E47).

9. CAP is itself an imperfect measure of liver fat: two operators agree with a concordance of 0.82 [12], and CAP correlates with MRI-PDFF at $r = 0.66$ [61], sharing about 44% of its variance with the reference measure.

Two causes dominate. First, about half of liver fat is genetic: in a prospective twin study the heritability of steatosis measured by MRI-PDFF was 0.52 (95% CI 0.31–0.73) [69]. Variants such as *PNPLA3* I148M change liver fat substantially at the same body size, and NHANES 2017–2023 contains no genotypes; lifelong diet, an accurate alcohol history and muscle mass are measured poorly or not at all. Two people with the same weight, waist and blood tests can therefore have very different livers. Second, the model is scored against a single CAP reading. With a reproducibility of about 0.82, a model that even knew about each patient's true liver state could not cross an R^2 of the order of 0.8. For routine data, the heritable share and the other unmeasured factors lower the attainable value further, to around 0.6. The in-scope result of 0.562 (0.593 in cross-validation) is close to that ceiling: the plateau is a property of the data, not of the model.

This is also why HepatoTwin relies on the calibrated probabilities of steatosis rather than on the exact CAP value. They discriminate well (AUC 0.88) and are well calibrated, and they outperform the established clinical scores on the same participants.

Table 3.10: Continuous liver-fat prediction from routine data.

Study	Population	Reference	Inputs	R^2
UK Biobank [65]	40,533 UK adults	MRI-PDFF	37 body and blood measures	0.37
UK Biobank [65]	9,723 UK adults	MRI-PDFF	Plasma metabolomics	0.34
UK Biobank [65]	46,000 UK adults	MRI-PDFF	DXA images (not routine)	0.63–0.64
Asaturyan et al. [64]	28,796 UK adults	MRI-PDFF	Fatty Liver Index alone	0.28–0.36
Asaturyan et al. [64]	As above	MRI-PDFF	12 variables incl. diagnoses	0.54
Petäjä et al. [67]	378 Finnish adults	^{1}H-MRS	Includes a specialist marker	0.44 / 0.49[c]
Liu H. et al. [53]	272, obesity clinic	CAP	BMI, neck, TG, C-peptide	0.63–0.65[d]
Haas et al. [66]	UK Biobank	MRI	BMI and 24 clinical factors	≈0.34[e]
Fatty Liver Index, this cohort	13,502 / 5,548	CAP	FLI (linear fit)	0.44 / 0.50
This study	15,300 NHANES	CAP	81 routine inputs	0.56 test; 0.59 CV

[c]Adjusted, discovery / validation. [d]Validation sample of a small, selected cohort with high CAP (median 320–335 dB/m against 251 in NHANES), which widens the outcome range and raises R^2. [e]Reported as $r = 0.58$. Kotronen et al. [62] and Chen et al. [68] report no R^2. This study: within the scope; 0.50 for all participants.

Table 3.11: NHANES steatosis classification models.

Study	Outcome	Reported	This study
Du et al. [49]	MASLD, CAP $\geq$ 280	AUC 0.815–0.842	0.879 (0.868)
Liu L. et al. [50]	CAP $\geq$ 302	AUC 0.859; FLI 0.838	0.888 (0.872)
Zhang et al. [51]	CAP $\geq$ 302	AUC 0.874	0.888 (0.872)
Boullion et al. [52]	CAP $\geq$ 238	Accuracy 79.6%	0.881 (0.857), CAP $\geq$ 248
Zhu et al. [63]	Asian, BMI $<$ 28	Macro AUC 0.874 (three grades)	0.799 test; 0.823 CV

This study: AUC within the scope, with all 3,060 test participants in brackets. The published studies cover adults of all ages, so the bracketed values are the closer comparison.

3.5 Comparison with Published Models

Published models fall into two groups: those predicting a continuous liver-fat value, which report R^2, and NHANES models classifying steatosis, which report AUC. No NHANES study reporting R^2 for CAP was found, so this work provides the first such benchmark.

Within its scope, the model's held-out R^2 of 0.56 (0.59 in cross-validation) is comparable to or above published routine-data models of liver fat (0.34–0.54; Table 3.10), including 0.37 in UK Biobank and 0.49 in validation for a spectroscopy equation that needs a specialist blood marker. Across all ages its R^2 of 0.50 still exceeds the UK Biobank routine-data model. Higher values come only from a small selected sample (0.65) or from imaging inputs (0.63–0.64). The studies differ in reference measure and population, so these values are indicative rather than directly comparable; the claim that can be made is "comparable to or above published routine-data models", not "state of the art".

In classification the model matches or exceeds every NHANES model (Table 3.11). The exception is lean Asian participants, where its AUC (0.80) is below the three-grade macro AUC of Zhu et al. (0.874), a metric that is not directly comparable and that rests on a training set of about 150,000 Chinese adults.

3.6 Summary

Fifty-one experiments raised R^2 from 0.491 for the first 14-input model to 0.562 on the held-out test set within the model's scope (ages 12 to 59, reliable reference examination), 0.593 in repeated cross-validation, and 0.578 on participants examined two to six years after the training data. Changes of algorithm, features, loss or tuning each moved R^2 by about 0.01 or less; the gains came from adding information (more NHANES variables and a second cycle) and from defining the population in which the model is used. The plateau near 0.56–0.59 is set by the data: about half of liver fat is heritable and unmeasured, and CAP is itself a noisy measure. At this level the model is comparable to or above published routine-data models, and its calibrated steatosis probabilities, which HepatoTwin uses, outperform the Fatty Liver Index and the Hepatic Steatosis Index.

The main limitations are that all validation is internal to NHANES; that the final settings were chosen with the help of the test set, which makes the test R^2 slightly optimistic; that patients aged 60 and over are outside the scope; and that the data are cross-sectional, so the model describes the current state and cannot by itself predict the response to a change in diet.

Chapter 4

Methodology

This chapter describes how HepatoTwin is built and how it is evaluated. Section 4.1 introduces the data: the NHANES FibroScan cohort, the patients' 24-hour dietary recalls, the recipe corpus and the external NHANES III cohort. Sections 4.2 to 4.7 mirror the different levels in Figure 1.1 namely the model of liver state and its safety criteria (Layer 2), the evidence engine (Layer 3), the recipe evaluation and prioritization engine (Layer 4), the scenario engine (Layer 5), the loop for updating the twin based on incoming data, and the layperson's report. Section 4.8 describes the software system and the patient and clinician interfaces (Layers 6 and 7). Section 4.9 sets out how each component is evaluated, including the in-silico evaluation of the twin on virtual patients. The experiments behind the liver-state model are reported separately in Chapter **??**; the results of every evaluation are given in Chapter efchap:results.

4.1 Data

4.1.1 NHANES FibroScan cohort

The twin is built on the US National Health and Nutrition Examination Survey (NHANES), which performed vibration-controlled transient elastography (FibroScan) on a nationally representative sample in its 2017–March 2020 and 2021–2023 cycles [13]. Both cycles use the same examination protocol and variable names. Participants with a complete examination (LUAXSTAT = 1) and an observed controlled attenuation parameter (CAP) form the cohort of 15,300 participants aged 12 to 80. A test set of 3,060 participants (20%, stratified by cycle, seed 42) was set aside before any model choice and is used only for evaluation; the remaining 12,240 form the training set. The split is saved and reused by every later component, so the same test participants are never seen in training anywhere in the system.

The liver-state model is intended for people aged 12 to 59, evaluated against reliable examinations (CAP interquartile range below 40 dB/m); this is the model's *scope* (Chapter **??**). The test set holds 1,236 in-scope participants, who are the held-out patients for every evaluation in this thesis.

4.1.2 Habitual diet from the 24-hour recall

Each NHANES participant reports everything eaten on the previous day in a 24-hour dietary recall. The day-1 nutrient totals of both cycles are joined with the USDA Food Patterns Equivalents (FPED), which give added sugars and food-group amounts. USDA has released FPED totals only for 2017–March 2020; for 2021–2023 they are derived with USDA's own method, from the individual foods and the 2017–2020 food-level FPED values (about 99.6% of the recalled energy has a matching food code). From the recall the twin computes the same composition measures it uses for recipes—energy, saturated fat and sugars as a share of energy, and fibre per 1,000 kcal—and the Healthy Eating Index 2020 [70].

A single day is a noisy measure of an individual's usual diet [44], and people under-report what they eat [43]. It is only in the event of NHANES judging the recall as reliable, and also if the participant indicates that the day was normal, that the recall may be considered (500–3,500 kcal for females, 800–4,000 kcal for males). Even in this case, the recall can affect the ranking of diet by not more than one level (Section 4.3). New users who are not NHANES participants type what they ate yesterday in plain words; each item is linked to a USDA SR Legacy food, converted to grams and summed into the same totals.

4.1.3 Recipe corpus

Recipes come from RecipeDB [26], which records each recipe's ingredients with their quantities, instructions, preparation time, servings, nutrition per serving, cuisine, diet labels and user ratings. A sample of 20,000 recipes was collected through the RecipeDB interface. Only main meals are candidates for recommendation: desserts, drinks, sauces, dressings and spreads are excluded by category and title, as are dishes with less than 200 kcal per serving. This leaves 10,108 candidate meals. Most (71%) are labelled "Other" rather than with a specific cuisine; the largest labelled cuisines are European (1,152), Asian (637), Mexican (308), Canadian (247) and American (228).

Every ingredient line is linked to a USDA SR Legacy food and its amount converted to grams, which allows two measures that RecipeDB does not provide: the degree of processing and the glycaemic load (Section 4.4).

4.1.4 NHANES III: external cohort with long-term follow-up

The external validation uses NHANES III (1988–1994), in which an archived gallbladder ultrasound was re-read for hepatic steatosis in 2009–2010 **nhanes3˙ultrasound**. Adults aged 20–59 with a gradable image form a cohort of 10,682 participants, 20.7% of whom had moderate or severe steatosis (the recode recommended by the National Center for Health Statistics). NHANES III examination and laboratory variables were mapped onto the model's inputs with unit conversions; 51 of the 81 inputs are available, and the rest are treated as missing, which the model handles natively. The public-use Linked Mortality File follows the participants to 31 December 2019 **nchs2022mortality**: 10,671 are linked, with 2,173 deaths over a median of 27.3 years. Its cause-of-death recode distin-

guishes heart disease, stroke and cancer but has no liver-disease category.

4.2 Layer 2: Liver-State Model and Clinical Safety Checks

4.2.1 Liver-state model

The liver twin's core is an XGBoost regressor [38] that estimates CAP from 81 routine inputs—body measurements, standard blood tests, blood pressure and a short questionnaire—and six derived scores. Two classifiers estimate the probability that CAP is at least 248 dB/m ("some liver fat", steatosis grade S1 or above) and at least 280 dB/m ("a lot of liver fat", grade S3) [11]; both are calibrated with isotonic regression [39]. The experiments that led to this model are reported in Chapter **??**.

The calibrated probabilities, not the CAP estimate, are the twin's primary state: with a held-out R^2 of 0.56 the individual CAP error (RMSE about 41 dB/m) is wider than the 32 dB/m between the S1 and S3 cut-offs, so a single CAP number would suggest more precision than the model has. Each probability is turned into a yes/no decision at the threshold that maximised the F1 score on the training data (0.432 for CAP $\geq$ 248 and 0.365 for CAP $\geq$ 280).

4.2.2 Uncertainty and explanation

Each patient's CAP estimate carries a 90% conformal prediction interval **vovk2005conformal**: five-fold out-of-fold predictions on the training set give one absolute error per in-scope training patient, and the interval is the estimate plus or minus their 90th percentile, with the finite-sample correction. Each calibrated probability carries a band, the 5th to 95th percentile of the probabilities given by isotonic calibrators refitted on 200 bootstrap resamples. SHAP values [40], computed with XGBoost's built-in TreeSHAP, split each patient's CAP estimate into the average prediction plus one contribution per input; they describe what drives the model's estimate, not causes that diet can change.

4.2.3 Fibrosis risk, liver label and safety notices

Diet advice must not replace the clinical care pathway. Three rule-based checks run alongside the model.

Fibrosis risk. FIB-4 = age $\times$ AST / (platelets $\times$ $\sqrt{\text{ALT}}$) **sterling2006fib4** is computed for adults and categorised by the EASL–EASD–EASO 2024 pathway [4]: below 1.3 low risk, 1.3–2.67 indeterminate (further testing advised) and above 2.67 high risk. A measured FibroScan liver stiffness, when available and reliable, replaces FIB-4 (below 8 kPa low, 8–12 kPa indeterminate, 12 kPa or more high). Because FIB-4 is less accurate under 35 [58], a low score in a young adult who is likely to have liver fat carries a note that elastography would help. A high risk withholds every recipe until the patient has had a clinical assessment.

Liver label. Following the 2023 multisociety nomenclature [3], steatotic liver disease with at least one cardiometabolic risk factor is labelled MASLD; with an alcohol intake of 140–350 g a week (women) or 210–420 g (men) MetALD; above that range alcohol-related liver disease; and without a risk factor cryptogenic. Adolescents are checked against the nomenclature's paediatric criteria. This is simply a research label that is used in the explanations and not a diagnosis. **Safety notices.** Those below 18 years of age, patients who have consumed alcohol at or above the MetALD level, on glucose lowering drugs, or those who are pregnant will get a note asking to consult a doctor before making any changes to the diet (Section 4.6).

4.3 Layer 3: Evidence Engine

4.3.1 Intervention evidence table

The evidence engine rests on a curated table of 25 intervention–outcome findings. Each row records the study, intervention, outcome, reported effect and direction, design, population, sample size, limitations and an explicit evidence-strength label: 4 rows are STRONG (for example, among patients who lost 5% or more of their weight in a 12-month prospective study with paired liver biopsies, 58% had resolution of steatohepatitis [5]), 11 MODERATE, 2 GUIDELINE RECOMMENDATION, 5 WEAK or INCONCLUSIVE and 3 OBSERVATIONAL ASSOCIATION. Observational associations are recorded but never used as effects.

4.3.2 From patient state to dietary priorities

Layer 2's decisions and routine measurements set explicit yes/no flags with published thresholds: probable steatosis and probable moderate-to-severe steatosis (from the calibrated probabilities); overweight and obesity (BMI 25 and 30 kg/m^2, or 23 and 27.5 for Asian patients); a raised waist (above 102 cm in men and 88 cm in women); raised glycaemia (HbA1c 5.7% or more); and dyslipidaemia (raised triglycerides or low HDL). Evidence rules then fire on these flags and produce up to six *priorities*, each with a level (low, medium or high) and the evidence rows that support it (Table 4.1). Every priority requires probable steatosis; the other flags decide which priorities apply and how high they rank. Priorities say which dietary objective matters more for this patient; they are not prescriptions and carry no numeric intake target.

When the patient has a reliable recall, a priority moves up one level if the patient's own intake misses the same reference value used for recipes, and down one level if it meets it.

4.3.3 Causal knowledge graph

A small hand-written causal graph of 16 links encodes the well-established direction of MASLD biology: dietary and activity levers act on body weight, insulin resistance and liver fat; liver fat drives liver injury and fibrosis; CAP, ALT and liver stiffness only measure them. Each link names its evidence rows. The graph confines the prioritization to the levers having a causal pathway towards

Table 4.1: Dietary priorities: when each applies (always together with probable steatosis), its default level, what raises it, and the recipe criterion it sets.

Priority	Also requires	Default	Raised to	Recipe criterion (reference)
Weight management	Excess weight	Medium	High: probable CAP ≥ 280, or obesity	Energy per serving $\leq$ 600 kcal
Diet quality	–	Medium	High: probable CAP ≥ 280	Fibre $\geq$ 14 g per 1,000 kcal
Limit saturated fat	–	Medium	High: dyslipidaemia	Saturated fat $\leq$ 10% of energy
Limit excess sugar	–	Low	Medium: raised glycaemia or excess weight	Sugars $\leq$ 10% of energy
Limit ultra-processed foods	–	Low	–	Processing FPro $\leq$ 0.5
Lower glycaemic load	Raised glycaemia	Low	–	Glycaemic load $\leq$ 20 per serving

liver fat, and therefore the criteria of any recipe never focuses on any downstream measurement such as ALT, and it provides the chain for every recommendation for the explanations (Section 4.4).

4.4 Layer 4: Recipe Evaluation and Ranking Engine

Layer 4, NutriTwin, is the core contribution of this thesis. It scores every candidate meal on five criteria, ranks first the meals that meet every evidence-linked reference value, and explains each recommendation.

4.4.1 Food preferences

At sign-up the patient states how they eat (anything, pescatarian, vegetarian, vegetarian without eggs, or vegan), any of the nine major food allergens named in US law plus gluten, any other allergy typed in, foods never eaten (pork, beef or alcohol), disliked and liked ingredients, the longest time they can spend cooking, and how many people they cook for. Diet, allergies, foods avoided, dislikes and cooking time are hard rules: a recipe that breaks one is never shown. A recipe must pass both RecipeDB's own diet label and a check of its ingredient words and food groups, because the labels contain errors (162 recipes labelled vegan list meat).

4.4.2 Criteria

Health relevance. Each priority maps to one recipe-composition criterion with a reference value from public dietary guidance (Table 4.1): energy per main meal **phe2018calories**, saturated fat **who2023satfat**, **dga2020**, sugars **who2015sugars** and fibre **dga2020**. Two criteria are computed

from the ingredients. The degree of processing (FPro, 0 to 1) is the energy-weighted mean of the ingredients' FPro, predicted by a re-implementation of FoodProX [77], a random forest that infers the NOVA processing class of a food from its nutrient profile. The glycaemic load per serving is the sum over ingredients of glycaemic index times available carbohydrate divided by 100 [78], with each carbohydrate food taking the median glycaemic index of its matching entries in the 2021 international tables. Both are left blank when too few ingredients can be linked. A criterion scores 100 at or past its reference value and falls linearly to 0 over a tolerance; the health score is the mean of the patient's criteria weighted by priority level.

Cuisine compatibility. A recipe labelled with the requested cuisine scores 100, a declared related cuisine 75 and any other label 0. Because 71% of candidates are unlabelled, these are judged on their ingredients: the typicality of a recipe for a cuisine is the mean share of that cuisine's recipes that use each pair of the recipe's ingredients, scaled so that the cuisine's own recipes at the 75th percentile score 100. Unlabelled recipes score at most 50, so they fill gaps without displacing correctly labelled ones.

Practical feasibility. The mean of three components, each 100 at the minimum and 0 at a screening limit: total time (60 minutes), number of ingredients (15) and number of instruction steps (12). The score is then reduced by 25% for each demanding technique named in the instructions (deep-frying, kneading or proving dough, overnight chilling, marinating, thermometer work, flambéing) and by 10% for each piece of special equipment.

Acceptability. RecipeDB's user rating, shrunk towards the corpus mean (4.91) with the weight of 20 ratings so that a recipe rated 5.0 by one user does not outrank one rated 4.9 by hundreds, and scaled from 4.5 (0) to 5.0 (100).

Personal fit. The closeness of a recipe to what the patient already eats and likes, with two parts averaged when both are known: the distance of the recipe from the patient's 24-hour recall on the patient's own priorities (meal energy, saturated fat, sugar and fibre, each in units of the criterion's tolerance, capped at 1), and one minus the largest ingredient overlap (Jaccard index) with a favourite dish or the liked ingredients. The personal fit is equal to 100 times (one minus this distance); if there is no reliable recall or favorites then it is unknown.

4.4.3 Health-gated ranking

The total score is the weighted average of the criteria which the recipe satisfies, where the weights are renormalized according to the criteria satisfied (Table 4.2). The weighted scoring alone allows such a trade-off. The ranking is, in this case, *health-gated*: recipes that fulfill all of the evidence-based reference values rank above the rest, and the weight score ranks recipes within these categories. Both processing and glycaemic load determine the score but are not health gated since the evidence behind them is less conclusive.

Inside the gated group every health score would be 100 and the patient's priority levels would no longer matter. There, half of the health component is instead the priority-weighted margin by which the recipe goes past the reference values, so a patient whose saturated-fat priority is high

is offered recipes further below the saturated-fat reference. Personal fit is applied only within the gated group, after ranking, so familiarity never outranks health.

Table 4.2: Criterion weights. Weights are renormalised over the criteria a recipe has; personal fit counts only for recipes that meet every reference value.

Criterion	Weight	Share when all are present
Health relevance	0.6	0.400
Cuisine compatibility	0.2	0.133
Practical feasibility	0.2	0.133
Acceptability	0.1	0.067
Personal fit	0.4	0.267

The personal-fit weight was chosen on the evaluation of Section 4.9: 0.4 made the top-ten lists of different patients clearly different while every recommended recipe still met every reference value (Chapter efchap:results). The other weights were set in the proposal and tested for sensitivity over a grid of 36 combinations.

4.4.4 Explanations, diversity and weekly plans

Each recommended recipe is explained by following its criteria back through the causal knowledge graph to liver fat, with the evidence rows and strength of each link, and ending with the cuisine and practicality context that kept the recipe appropriate. Every statement is built from the ranking engine's own output.

A seven-day plan of two main meals a day is drawn greedily from the gated recipes, practical ones first, after re-ranking by maximal marginal relevance [76] ($\lambda = 0.9$) so that near-duplicates do not fill the list. The solution needs to meet the constraints on a weekly basis. This involves meeting the requirements of each meal and each pair of daily meals based on the reference values of the patient, the recipe appearing twice or not at all in the week and not consecutively, there being a difference in protein families in daily meals, and the absence of a dominant protein family in the week. If the cuisine cannot fill the week, a favourite may appear twice, and then recipes from the most similar cuisines (by ingredient profile) are borrowed and labelled as such. A shopping list adds up one serving of each planned meal.

4.5 Layer 5: Scenario Engine

The patient chooses between two scenarios. *Scenario A* takes the recipes in the health-gated order, the healthiest available. *Scenario B* takes recipes from the same safe pool ordered by closeness to what the patient eats now, so the change is smaller. For each criterion the engine compares the scenario's mean composition with the patient's own recall and reports whether it improves, worsens or does not change meaningfully, each direction carrying the evidence rows of its pathway: calorie restriction (and, if sustained to 5% weight loss, the STRONG weight-loss evidence [5]); saturated fat

[9]; sugar, labelled MODERATE only when the scenario also lowers energy, because sugar harms the liver when it adds surplus energy but not when exchanged isocalorically [10]; and fibre as a marker of a Mediterranean-style pattern [7]. The scenario's label is the strongest among the criteria that improve, and every scenario is labelled as indirect evidence, since one recipe is not a dietary pattern.

The scenario engine never asks the Layer 2 model to predict an intervention's effect, because the model was trained on a cross-sectional survey, not on interventions. Two intermediate steps between the scenarios are used only by the updating loop, which can move a patient gradually from Scenario B towards Scenario A as they settle in, or back when they struggle (Section 4.6).

4.6 Updating the Twin: Plan, Check-ins and Re-assessment

4.6.1 The plan

Patients cannot be expected to know how large a change to make, so the twin suggests a plan at three paces, taken from the trial prescriptions: gentle (half the Diabetes Prevention Program calorie cut for the patient's weight band and 90 minutes of walking a week), standard (the DPP cut of 500–1,000 kcal a day and 150 minutes) [85] and ambitious (the Look AHEAD calorie goal and 175 minutes) [86]. Walking adds (MET -1) $\times$ weight $\times$ hours of energy use at 3.8 MET. Patients under 18 receive no calorie cut. A plan with a safety notice waits for a clinician's approval, and only a clinician can set custom numbers.

4.6.2 Predict, observe, compare, adjust

At each monthly check-in the patient enters their weight and how many of the planned meals they cooked and how much they walked. The twin then runs one cycle of a predict–observe–compare–adjust loop.

Predict. The forecast follows Hall's published rule [80]: a sustained change in intake moves weight by about 1 kg per 100 kJ a day in the end, half of it in about a year. The plan's full effect is scaled by a personal response factor $k = a \times r$, the product of adherence a (the share of the plan followed, from the patient's answers) and the body's response r (1 for the textbook response).

Compare. The 90% range of the forecast combines the uncertainty in k with ordinary weight drift, set from the spread of weight change in Look AHEAD usual care and assumed to grow like a random walk. Progress is also judged against the plan from the start: from month 1.5, a patient who has lost less than half of the loss the plan typically gives is *behind plan.* It can be seen from the patient's responses that if the adherence level is low, then the adherence to the regimen has not taken place, and vice versa. *Adjust.* Response variable r is reassessed by Bayesian linear regression using the changes in weight between visits from a weakly informative prior ($r = 1.0 \pm 0.7$), with older values discounted due to adherence drift. The projection at 3, 6 and 12 months uses what has been learned, and the liver projection re-runs Layer 2 with the projected weight, BMI and waist. A

plan changed by the clinician part-way through is handled as a dated schedule of energy changes, so each month is judged against the plan in force at that time.

4.6.3 Tips and alerts

Every message follows from one rule on the twin's output. The patient that did not adhere to the plan will be provided with more familiar recipes, while the one that adhered to the plan will be advised to continue following it. After two consecutive follow-ups of patients on the plan, the doctor will receive a notification of the points to consider: intake greater than stated, regular fluctuations in weight, and medical issues like newly prescribed medication or hyperthyroidism. The alert does not claim the patient's body is slow: the in-silico evaluation showed that this cannot be told apart from weight drift within a year (Chapter efchap:results). A patient on plan with good adherence at two check-ins in a row is offered one step towards the healthiest recipes. Clinicians also see alerts for plans awaiting approval, indeterminate or high fibrosis risk, overdue check-ins and missing blood tests.

4.7 Plain-Language Summary

The patient reads a short summary of the twin's output. It is produced solely based on the facts computed by the twin, with all numbers being rounded up as indicated: the probabilities of a few and a high amount of fat in the liver (never as 0% or 100%), a clear verdict based on the twin's binary decision, the liver classification, the outcome of fibrosis using published thresholds and the sources for those thresholds (with an indication where the score falls within 10% of the threshold), any warnings about risks, and the importance of criteria in clear language with the number of studies supporting each, and the top recipes. Evidence codes, the CAP estimate and its interval are kept for the clinician's view. Notices about changing diet appear only when a diet change is suggested.

A template writes the summary, and a language model can be plugged in instead. Whatever writes it, an automatic check confirms that every number and every evidence citation in the text appears in the twin's facts; if not, the template summary is shown and the reason recorded.

4.8 System and Interfaces

The twin's Layers 2–5 are wrapped in an HTTP backend (FastAPI) with an SQLite database that keeps each patient's dated visits, plans, check-ins and twin runs, so later visits can be compared with what the twin expected. Patients sign in with a one-time code and see only their own record; clinicians see the patients of their clinic. One twin run has two views. The patient's view holds the summary, the chances of liver fat, the safety check, the priorities, the recipes with their reasons and the two scenarios, but no CAP estimate. The clinician's view holds the full output, including the CAP estimate, its 90% interval and the SHAP drivers.

The dashboard (Layer 7) presents this journey: a short sign-up, the patient's results, the choice of pace and recipe style, a "Today" screen with one next step and the day's meals, the weekly plan and shopping list, monthly check-ins, and progress against the plan; and for the clinician, a list of patients ordered by alerts and a patient view in which plans are approved or changed. The liver visualisation (Layer 6) colours a three-dimensional liver by the steatosis state from the calibrated probabilities, with a timeline slider for today and three, six and twelve months, and labels every future state as a simulated projection, not a clinical forecast.

4.9 Evaluation Design

Each component is evaluated against appropriate baselines on data it was not built on. Unless stated, intervals are 95% bootstrap percentile intervals over participants, and results are unweighted.

4.9.1 Liver-state model and safety checks

The model is evaluated on the in-scope test participants by R^2 and RMSE for CAP, and by ROC-AUC, Brier score and calibration for the two classifiers, against the Fatty Liver Index [47] and the Hepatic Steatosis Index [48] on the same participants (Chapter **??**). The conformal intervals are checked by their empirical coverage on the test set (target 90%). FIB-4 is evaluated against liver stiffness measured in the same FibroScan examination, with 8 kPa to rule out and 12 kPa to rule in advanced fibrosis.

4.9.2 Fairness audit

The calibrated classifiers are scored separately by sex, race and ethnicity, and age group (12–17, 18–39, 40–59) on the in-scope test participants: ROC-AUC, Brier score, and the mean predicted probability against the observed share (calibration in the large), with 1,000 bootstrap resamples within each group. Groups with fewer than 30 participants are not reported. The share of recommended recipes meeting each ranking criterion is compared across the same groups.

4.9.3 External and long-term validation on NHANES III

On the NHANES III cohort, the model's probabilities are compared with the ultrasound grade (moderate or severe against normal or mild) by ROC-AUC, alongside the Fatty Liver Index (on participants with GGT), the Hepatic Steatosis Index, BMI and waist computed from the same record. This is a different reference standard 30 years earlier, so it tests whether the model ranks people correctly, not whether it reproduces CAP. For mortality, participants are divided into quarters of the probability of CAP ≥ 280, and death rates per 1,000 person-years are compared for all causes, cardiovascular causes (heart disease or stroke) and cancer. Rate ratios adjusted for age and sex come from Poisson regression with the logarithm of person-years as offset, fitted by iteratively

reweighted least squares, with Wald intervals. The ultrasound grade and the FIB-4 category are analysed the same way.

4.9.4 Recipe ranking

The ranking engine is evaluated on 300 randomly chosen in-scope test participants with at least one recipe criterion. NHANES records no cuisine preference, so each participant is given a cuisine at random among those with at least 100 candidate recipes. The main measure is the share of each top-ten list that meets all three criteria at once: every one of the patient's health reference values, the patient's cuisine, and the practical limits (60 minutes, 15 ingredients, 12 steps). The proposed ranking is compared with the same ranking without personal fit, without acceptability and with diversity re-ranking; with weighted scoring without the health gate; with health-only and nutrition-only rankings; and with random selection. Personalisation is measured by the number of distinct top-ten lists and the mean overlap of the top ten between patients of the same cuisine, with and without the 24-hour recall. Weekly plans are checked against every weekly constraint, and the sensitivity of the ranking to its weights is measured over 36 weight combinations.

4.9.5 Plain-language summaries

For 300 in-scope test participants the twin runs end to end. The summaries are scored on faithfulness (every number and citation present in the twin's facts) and on readability (Flesch Reading Ease and Flesch–Kincaid grade **flesch1948readability**). The faithfulness check itself is tested by planting errors in each summary as a language model might make them—a number changed, a number added, a citation invented, which should all be caught; a harmless rewording, which should pass; and a wrong word with no number, which measures the check's blind spot. A random sample of ten summaries was read by the project team, each beside the words its rules should produce.

4.9.6 In-silico evaluation of the updating twin

NHANES examined each participant once, so the twin's updating and projections cannot be tested on real follow-up. Following the design of in-silico trials [54] and plasmode simulation [57], real test patients provide month 0 and their next twelve months are simulated; this addresses the verification part of digital twin validation [14].

Simulator. Each of the 1,236 in-scope test patients follows usual diet, Scenario A (−500 kcal a day) and Scenario B (−250 kcal a day) for twelve months. A person-level adherence pattern (steady, slipping, regaining or dropping out) scales the prescribed change. Monthly weight follows Hall's model [80], scaled by each person's body fat, with a random weight drift independent of the diet. Waist, blood markers and blood pressure change with weight loss by published effect sizes [83]; liver fat falls with weight loss and further while intake is reduced [60]; CAP follows liver fat, and liver enzymes follow the change in liver fat **wong2013community**, [59]. Each person's biology is drawn once, so the scenarios differ only in diet. The simulator never uses the Layer 2 model, so the twin

is not checked against its own output.

Observation layer. The twin never sees the simulated truth. It sees what a clinic would record: monthly weights and food logs, blood tests every three months with within-person and laboratory variation from the EFLM database [84], CAP at months 0, 6 and 12 with measurement error, food logs that under-report intake as in the OPEN study [43], and missed visits that become more likely when a patient's weight has risen.

Calibration. The settings that no trial reports directly (adherence patterns and how they fade, response spread, weight drift) were tuned by differential evolution so that virtual patients resembling each trial's participants, drawn from the training set, reproduce published outcomes: Look AHEAD at one year [86], the DPP at 24 weeks [85] and Vilar-Gomez et al. at one year [5]. During each programme's intensive-contact phase a slipping participant keeps full effort. Later outcomes (Look AHEAD at eight years [81] and the DPP at about three years) were held back to check the tuned simulator, and first-year regain in Look AHEAD **neiberg2012patterns** was used only to choose how many participants stop the diet completely. HepatoTwin's scenarios use the adherence fitted to Vilar-Gomez et al., routine advice to patients with fatty liver.

Measures. The twin runs month by month on the observed data with the patient's answers derived from the true adherence (honest answers) and, as a robustness check, inflated by each person's under-reporting share, or absent. The performance of the model is judged by: predictions of the weight at months 3 and 6, 3 and 6 months ahead, compared to the textbook pattern without learning and to the last carried forward weight (mean absolute error and coverage of the 90% range); how the prediction error varies with the learning process from month 1 to month 9; the reasons given for the slow learning and warnings, compared to actual adherence and body reaction; how many months it takes to detect a patient falling behind and liver fat falling below 280 dB/m; the probability of CAP $\geq$ 280 at months 0, 6 and 12 against the true CAP (AUC, Brier score, calibration), and its month-6 forecast of month 12; robustness to more missed visits, scale noise, over-reported answers and missing answers; and whether its projections order Scenario A, Scenario B and usual diet as the truth does. These results verify that the twin can update, track and adapt; they do not show that it predicts real patients, and almost all calibration trials were in adults.

Chapter 5

Results

Bibliography

[1] Z. M. Younossi, P. Golabi, J. M. Paik, A. Henry, C. Van Dongen, and L. Henry, "The global epidemiology of nonalcoholic fatty liver disease (NAFLD) and nonalcoholic steatohepatitis (NASH): A systematic review," *Hepatology*, vol. 77, no. 4, pp. 1335–1347, 2023. DOI: 10.1097/HEP.0000000000000004

[2] K. Riazi et al., "The prevalence and incidence of NAFLD worldwide: A systematic review and meta-analysis," *The Lancet Gastroenterology & Hepatology*, vol. 7, no. 9, pp. 851–861, 2022. DOI: 10.1016/S2468-1253(22)00165-0

[3] M. E. Rinella, J. V. Lazarus, V. Ratziu, S. M. Francque, A. J. Sanyal, F. Kanwal, et al., "A multisociety Delphi consensus statement on new fatty liver disease nomenclature," *Hepatology*, vol. 78, no. 6, pp. 1966–1986, 2023.

[4] European Association for the Study of the Liver (EASL), European Association for the Study of Diabetes (EASD), and European Association for the Study of Obesity (EASO), "EASL-EASD-EASO clinical practice guidelines on the management of metabolic dysfunction-associated steatotic liver disease (MASLD)," *Journal of Hepatology*, vol. 81, no. 3, pp. 492–542, 2024. DOI: 10.1016/j.jhep.2024.04.031

[5] E. Vilar-Gomez et al., "Weight loss through lifestyle modification significantly reduces features of nonalcoholic steatohepatitis," *Gastroenterology*, vol. 149, no. 2, pp. 367–378, 2015. DOI: 10.1053/j.gastro.2015.04.005

[6] A. A. Sangouni et al., "Effect of Mediterranean diet on liver enzymes: A systematic review and meta-analysis of randomised controlled trials," *British Journal of Nutrition*, vol. 128, no. 7, pp. 1231–1239, 2022. DOI: 10.1017/S0007114521002270

[7] L. Haigh, C. Kirk, K. El Gendy, J. Gallacher, L. Errington, et al., "The effectiveness and acceptability of Mediterranean diet and calorie restriction in non-alcoholic fatty liver disease (NAFLD): A systematic review and meta-analysis," *Clinical Nutrition*, vol. 41, no. 9, pp. 1913–1931, 2022. DOI: 10.1016/j.clnu.2022.06.037

[8] Y. Xiong et al., "Effects of Mediterranean diet and low-fat diet on liver enzymes and liver fat content of NAFLD: A meta-analysis of randomized controlled trials," *Food & Function*, vol. 15, no. 16, pp. 8248–8257, 2024. DOI: 10.1039/D4FO01461H

[9] P. K. Luukkonen et al., "Saturated fat is more metabolically harmful for the human liver than unsaturated fat or simple sugars," *Diabetes Care*, vol. 41, no. 8, pp. 1732–1739, 2018. DOI: 10.2337/dc18-0071

[10] S. Chiu et al., "Effect of fructose on markers of non-alcoholic fatty liver disease (NAFLD): A systematic review and meta-analysis of controlled feeding trials," *European Journal of Clinical Nutrition*, vol. 68, no. 4, pp. 416–423, 2014. DOI: 10.1038/ejcn.2014.8

[11] T. Karlas et al., "Individual patient data meta-analysis of controlled attenuation parameter (CAP) technology for assessing steatosis," *Journal of Hepatology*, vol. 66, no. 5, pp. 1022–1030, 2017. DOI: 10.1016/j.jhep.2016.12.022

[12] G. Ferraioli, "Quantitative assessment of liver steatosis using ultrasound controlled attenuation parameter (Echosens)," *Journal of Medical Ultrasonics*, vol. 48, pp. 489–495, 2021. DOI: 10.1007/s10396-021-01106-1

[13] National Center for Health Statistics, *National health and nutrition examination survey: Liver ultrasound transient elastography (LUX) data documentation, 2017–march 2020 and august 2021–august 2023*, Centers for Disease Control and Prevention, Hyattsville, MD, 2024.

[14] National Academies of Sciences, Engineering, and Medicine, *Foundational Research Gaps and Future Directions for Digital Twins*. Washington, DC: The National Academies Press, 2024. DOI: 10.17226/26894

[15] R. Laubenbacher, B. Mehrad, I. Shmulevich, and N. Trayanova, "Digital twins in medicine," *Nature Computational Science*, vol. 4, no. 3, pp. 184–191, 2024. DOI: 10.1038/s43588-024-00607-6

[16] E. Katsoulakis et al., "Digital twins for health: A scoping review," *npj Digital Medicine*, vol. 7, no. 1, p. 77, 2024. DOI: 10.1038/s41746-024-01073-0

[17] C. Su, P. Wang, N. Foo, and D. Ho, "Optimizing metabolic health with digital twins," *npj Aging*, vol. 11, no. 1, p. 20, 2025. DOI: 10.1038/s41514-025-00211-6

[18] M. D. Shen, S. B. Chen, and X. D. Ding, "The effectiveness of digital twins in promoting precision health across the entire population: A systematic review," *npj Digital Medicine*, vol. 7, no. 1, p. 145, 2024. DOI: 10.1038/s41746-024-01146-0

[19] D. Zeevi et al., "Personalized nutrition by prediction of glycemic responses," *Cell*, vol. 163, no. 5, pp. 1079–1094, 2015. DOI: 10.1016/j.cell.2015.11.001

[20] K. Gkouskou, I. Vlastos, P. Karkalousos, D. Chaniotis, D. Sanoudou, and A. G. Eliopoulos, "The "virtual digital twins" concept in precision nutrition," *Advances in Nutrition*, vol. 11, no. 6, pp. 1405–1413, 2020. DOI: 10.1093/advances/nmaa089

[21] S. Joshi, P. Shamanna, M. Dharmalingam, A. Vadavi, A. Keshavamurthy, L. Shah, et al., "Digital twin-enabled personalized nutrition improves metabolic dysfunction-associated fatty liver disease in type 2 diabetes: Results of a 1-year randomized controlled study," *Endocrine Practice*, vol. 29, no. 12, pp. 960–970, 2023. DOI: 10.1016/j.eprac.2023.08.016

[22] P. Shamanna et al., "Personalized nutrition in type 2 diabetes remission: Application of digital twin technology for predictive glycemic control," *Frontiers in Endocrinology*, vol. 15, p. 1 485 464, 2024. DOI: 10.3389/fendo.2024.1485464

[23] A. Nobel, J. Matos, H. Wang, and H. Fang, "iDT-diet: An intelligent digital twin model for diet-influenced biomarker trajectories (student abstract)," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 2026, pp. 41 334–41 336. DOI: 10.1609/aaai.v40i48.42260

[24] *ARTEMIS: Accelerating the translation of virtual twins towards a personalised management of steatotic liver patients*, Horizon Europe grant 101136299; ClinicalTrials.gov NCT07430501, 2024.

[25] G. Bagler and M. Goel, "Computational gastronomy: Capturing culinary creativity by making food computable," *npj Systems Biology and Applications*, vol. 10, no. 1, p. 72, 2024. DOI: 10.1038/s41540-024-00399-5

[26] D. Batra et al., "RecipeDB: A resource for exploring recipes," *Database*, vol. 2020, baaa077, 2020. DOI: 10.1093/database/baaa077

[27] N. Garg et al., "FlavorDB: A database of flavor molecules," *Nucleic Acids Research*, vol. 46, no. D1, pp. D1210–D1216, 2018. DOI: 10.1093/nar/gkx957

[28] N. K. Rakhi, R. Tuwani, J. Mukherjee, and G. Bagler, "Data-driven analysis of biomedical literature suggests broad-spectrum benefits of culinary herbs and spices," *PLoS ONE*, vol. 13, no. 5, e0198030, 2018. DOI: 10.1371/journal.pone.0198030

[29] O. Sahin and G. Aytekin-Sahin, "Open-source multi-objective optimization software for menu planning," *Expert Systems with Applications*, vol. 252, p. 124 213, 2024. DOI: 10.1016/j.eswa.2024.124213

[30] C. Trattner and D. Elsweiler, *Food recommender systems: Important contributions, challenges and future research directions*, arXiv preprint arXiv:1711.02760, 2017.

[31] S. Haussmann, O. Seneviratne, Y. Chen, et al., "FoodKG: A semantics-driven knowledge graph for food recommendation," in *The Semantic Web – ISWC 2019*, ser. Lecture Notes in Computer Science, vol. 11779, Springer, 2019.

[32] Y. Chen, A. Subburathinam, C.-H. Chen, and M. J. Zaki, "Personalized food recommendation as constrained question answering over a large-scale food knowledge graph," in *Proceedings of the 14th ACM International Conference on Web Search and Data Mining (WSDM)*, 2021.

[33] D. Li, M. J. Zaki, and C.-h. Chen, "Health-guided recipe recommendation over knowledge graphs," *Journal of Web Semantics*, p. 100 743, 2022. DOI: 10.1016/j.websem.2022.100743

[34] S. S. Shirai, O. Seneviratne, M. E. Gordon, C.-H. Chen, and D. L. McGuinness, "Identifying ingredient substitutions using a knowledge graph of food," *Frontiers in Artificial Intelligence*, vol. 3, p. 621 766, 2021. DOI: 10.3389/frai.2020.621766

[35] H. Kim, R. Venkataramanan, and A. Sheth, *A survey on food ingredient substitutions*, arXiv preprint arXiv:2501.01958, 2025.

[36] L. Rita, J. Southern, I. Laponogov, K. Higgins, et al., *Optimizing ingredient substitution using large language models to enhance phytochemical content in recipes*, arXiv preprint arXiv:2409.08792, 2024.

[37] Y. Cao et al., "Cultural adaptation of recipes," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 80–99, 2024. DOI: 10.1162/tacl_a_00634

[38] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[39] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002, pp. 694–699.

[40] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30*, 2017.

[41] J. G. Stine et al., "Exercise training is associated with treatment response in liver fat content by magnetic resonance imaging independent of clinically significant body weight loss in patients with nonalcoholic fatty liver disease: A systematic review and meta-analysis," *American Journal of Gastroenterology*, vol. 118, no. 7, pp. 1204–1213, 2023. DOI: 10.14309/ajg.0000000000002098

[42] U. M. Stern, G. Wagenpfeil, S. N. Weber, S. A. Alqahtani, M. Michel, J. M. Schattenberg, et al., "The impact of dietary interventions on liver health biomarkers in individuals with metabolic dysfunction-associated steatotic liver disease: A systematic review and meta-analysis of randomized controlled trials," *European Journal of Nutrition*, vol. 65, no. 2, p. 58, 2026. DOI: 10.1007/s00394-025-03870-z

[43] A. F. Subar, V. Kipnis, R. P. Troiano, D. Midthune, D. A. Schoeller, S. Bingham, et al., "Using intake biomarkers to evaluate the extent of dietary misreporting in a large sample of adults: The OPEN study," *American Journal of Epidemiology*, vol. 158, no. 1, pp. 1–13, 2003.

[44] K. W. Dodd, P. M. Guenther, L. S. Freedman, A. F. Subar, V. Kipnis, D. Midthune, et al., "Statistical methods for estimating usual intake of nutrients and foods: A review of the theory," *Journal of the American Dietetic Association*, vol. 106, no. 10, pp. 1640–1650, 2006. DOI: 10.1016/j.jada.2006.07.011

[45] J. A. Tooze, D. Midthune, K. W. Dodd, L. S. Freedman, S. M. Krebs-Smith, A. F. Subar, et al., "A new statistical method for estimating the usual intake of episodically consumed foods with application to their distribution," *Journal of the American Dietetic Association*, vol. 106, no. 10, pp. 1575–1587, 2006. DOI: 10.1016/j.jada.2006.07.003

[46] E. Kalnay, *Atmospheric Modeling, Data Assimilation and Predictability*. Cambridge: Cambridge University Press, 2003.

[47] G. Bedogni et al., "The fatty liver index: A simple and accurate predictor of hepatic steatosis in the general population," *BMC Gastroenterology*, vol. 6, p. 33, 2006.

[48] J.-H. Lee et al., "Hepatic steatosis index: A simple screening tool reflecting nonalcoholic fatty liver disease," *Digestive and Liver Disease*, vol. 42, no. 7, pp. 503–508, 2010.

[49] S. Du, H. Yu, J. Du, and Y. Xu, "Machine learning models to predict metabolic dysfunction-associated steatotic liver disease (MASLD) with simple anthropometric and biochemical variables: A cross-sectional study in US population," *International Journal of Hepatology*, vol. 2026, no. 1, p. 8 221 645, 2026. DOI: 10.1155/ijh/8221645

[50] L. Liu et al., "Automated machine learning models for nonalcoholic fatty liver disease assessed by controlled attenuation parameter from the NHANES 2017–2020," *Digital Health*, vol. 10, p. 20 552 076 241 272 535, 2024. DOI: 10.1177/20552076241272535

[51] Y. Zhang, X. Liu, X. Zhang, Y. Fei, and X. Li, "Machine learning-based prediction of metabolic dysfunction-associated steatotic liver disease using National Health and Nutrition Examination Survey (NHANES) data," *PLoS ONE*, vol. 20, no. 11, e0335656, 2025. DOI: 10.1371/journal.pone.0335656

[52] J. Boullion et al., "Machine learning–based biomarker identification for early diagnosis of metabolic dysfunction–associated steatotic liver disease," *The Journal of Clinical Endocrinology & Metabolism*, vol. 110, no. 11, e3866–e3877, 2025. DOI: 10.1210/clinem/dgaf111

[53] H. Liu et al., "Simple surrogate equations to predict controlled attenuation parameter values for screening non-alcoholic fatty liver disease in a Chinese population," *Frontiers in Medicine*, vol. 9, p. 894 895, 2022. DOI: 10.3389/fmed.2022.894895

[54] F. Pappalardo, G. Russo, F. M. Tshinanu, and M. Viceconti, "In silico clinical trials: Concepts and early adoptions," *Briefings in Bioinformatics*, vol. 20, no. 5, pp. 1699–1708, 2019. DOI: 10.1093/bib/bby043

[55] B. P. Kovatchev, M. Breton, C. Dalla Man, and C. Cobelli, "In silico preclinical trials: A proof of concept in closed-loop control of type 1 diabetes," *Journal of Diabetes Science and Technology*, vol. 3, no. 1, pp. 44–55, 2009. DOI: 10.1177/193229680900300106

[56] J. Walonoski, M. Kramer, J. Nichols, et al., "Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record," *Journal of the American Medical Informatics Association*, vol. 25, no. 3, pp. 230–238, 2018.

[57] J. M. Franklin, S. Schneeweiss, J. M. Polinski, and J. A. Rassen, "Plasmode simulation for the evaluation of pharmacoepidemiologic methods in complex healthcare databases," *Computational Statistics & Data Analysis*, vol. 72, pp. 219–226, 2014.

[58] M. E. Rinella, B. A. Neuschwander-Tetri, M. S. Siddiqui, M. F. Abdelmalek, et al., "AASLD practice guidance on the clinical assessment and management of nonalcoholic fatty liver disease," *Hepatology*, vol. 77, no. 5, pp. 1797–1835, 2023. DOI: 10.1097/HEP.0000000000000323

[59] D. A. Koutoukidis, N. M. Astbury, K. E. Tudor, E. Morris, J. A. Henry, et al., "Association of weight loss interventions with changes in biomarkers of nonalcoholic fatty liver disease: A systematic review and meta-analysis," *JAMA Internal Medicine*, vol. 179, no. 9, pp. 1262–1271, 2019.

[60] E. L. Lim, K. G. Hollingsworth, B. S. Aribisala, M. J. Chen, J. C. Mathers, and R. Taylor, "Reversal of type 2 diabetes: Normalisation of beta cell function in association with decreased pancreas and liver triacylglycerol," *Diabetologia*, vol. 54, no. 10, pp. 2506–2514, 2011.

[61] S. B. Ahn, D. W. Jun, B.-K. Kang, et al., "Optimal cutoff value for assessing changes in intrahepatic fat amount by using the controlled attenuation parameter in a longitudinal setting," *Medicine*, vol. 97, no. 50, e13636, 2018. DOI: 10.1097/MD.0000000000013636

[62] A. Kotronen, M. Peltonen, A. Hakkarainen, K. Sevastianova, R. Bergholm, et al., "Prediction of non-alcoholic fatty liver disease and liver fat using metabolic and genetic factors," *Gastroenterology*, vol. 137, no. 3, pp. 865–872, 2009. DOI: 10.1053/j.gastro.2009.06.005

[63] Y. Zhu et al., "Hepatic steatosis severity prediction in nonobese individuals: Machine learning model development and validation," *Journal of Medical Internet Research*, vol. 28, e82529, 2026. DOI: 10.2196/82529

[64] H. A. Asaturyan, N. Basty, M. Thanaj, B. Whitcher, E. L. Thomas, and J. D. Bell, "Improving the accuracy of fatty liver index to reflect liver fat content with predictive regression modelling," *PLoS ONE*, vol. 17, no. 9, e0273171, 2022. DOI: 10.1371/journal.pone.0273171

[65] *Machine learning across multiple imaging and biomarker modalities in the UK Biobank improves genetic discovery for liver fat accumulation*, medRxiv preprint, 2024. DOI: 10.1101/2024.01.06.24300923

[66] M. E. Haas, J. P. Pirruccello, S. N. Friedman, et al., "Machine learning enables new insights into genetic contributions to liver fat accumulation," *Cell Genomics*, vol. 1, no. 3, p. 100 066, 2021. DOI: 10.1016/j.xgen.2021.100066

[67] E. M. Petäjä et al., "Phosphorylated IGFBP-1 as a non-invasive predictor of liver fat in NAFLD," *Scientific Reports*, vol. 6, p. 24 740, 2016. DOI: 10.1038/srep24740

[68] J. Chen, J. Yue, J. Fu, et al., "A prediction model of liver fat fraction and presence of non-alcoholic fatty liver disease (NAFLD) among patients with overweight or obesity," *Endocrine Journal*, vol. 70, pp. 977–985, 2023. DOI: 10.1507/endocrj.EJ23-0227

[69] R. Loomba, N. Schork, C.-H. Chen, et al., "Heritability of hepatic fibrosis and steatosis based on a prospective twin study," *Gastroenterology*, vol. 149, no. 7, pp. 1784–1793, 2015. DOI: 10.1053/j.gastro.2015.08.011

[70] M. M. Shams-White, T. E. Pannucci, J. L. Lerman, K. A. Herrick, et al., "Healthy eating index-2020: Review and update process to reflect the dietary guidelines for americans, 2020–2025," *Journal of the Academy of Nutrition and Dietetics*, vol. 123, no. 9, pp. 1280–1288, 2023. DOI: 10.1016/j.jand.2023.05.015

[71] A.-K. Nitschke, C. Brandl, F. Egersdörfer, M. Görtz, M. Hohenfellner, and M. Weidemüller, "Design for a digital twin in clinical patient care," *npj Health Systems*, vol. 3, p. 11, 2026. DOI: 10.1038/s44401-025-00060-1

[72] L. Nye, *Digital twins for patient care via knowledge graphs and closed-form continuous-time liquid neural networks*, arXiv preprint arXiv:2307.04772, 2023.

[73] N. L. B. Freeman, Y. Zou, K. Lee, C. Yu, et al., *The fidelity and feedback traps: The case for health digital twins as modular evolving causal systems*, arXiv preprint arXiv:2607.27192, 2026.

[74] O. Laudy, *The digital twin counterfactual framework: A validation architecture for simulated potential outcomes*, arXiv preprint arXiv:2604.01325, 2026.

[75] J. Loesch, L. Meeckers, I. van Lier, A. de Boer, M. Dumontier, and R. Celebi, "Automated identification of food substitutions using knowledge graph embeddings," in *CEUR Workshop Proceedings*, 2022, pp. 19–28.

[76] J. Carbonell and J. Goldstein, "The use of MMR, diversity-based reranking for reordering documents and producing summaries," in *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, 1998, pp. 335–336.

[77] G. Menichetti, B. Ravandi, D. Mozaffarian, and A.-L. Barabási, "Machine learning prediction of the degree of food processing," *Nature Communications*, vol. 14, no. 1, p. 2312, 2023. DOI: 10.1038/s41467-023-37457-1

[78] F. S. Atkinson, J. C. Brand-Miller, K. Foster-Powell, A. E. Buyken, and J. Goletzke, "International tables of glycemic index and glycemic load values 2021: A systematic review," *American Journal of Clinical Nutrition*, vol. 114, no. 5, pp. 1625–1632, 2021.

[79] C. Dalla Man, F. Micheletto, D. Lv, M. Breton, B. Kovatchev, and C. Cobelli, "The UVA/PADOVA type 1 diabetes simulator: New features," *Journal of Diabetes Science and Technology*, vol. 8, no. 1, pp. 26–34, 2014. DOI: 10.1177/1932296813514502

[80] K. D. Hall et al., "Quantification of the effect of energy imbalance on bodyweight," *The Lancet*, vol. 378, no. 9793, pp. 826–837, 2011.

[81] Look AHEAD Research Group, "Eight-year weight losses with an intensive lifestyle intervention: The Look AHEAD study," *Obesity*, vol. 22, no. 1, pp. 5–13, 2014.

[82] Diabetes Prevention Program Research Group, "Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin," *New England Journal of Medicine*, vol. 346, no. 6, pp. 393–403, 2002.

[83] R. R. Wing, W. Lang, T. A. Wadden, M. Safford, W. C. Knowler, A. G. Bertoni, et al., "Benefits of modest weight loss in improving cardiovascular risk factors in overweight and obese individuals with type 2 diabetes," *Diabetes Care*, vol. 34, no. 7, pp. 1481–1486, 2011.

[84] European Federation of Clinical Chemistry and Laboratory Medicine, *EFLM biological variation database*, https://biologicalvariation.eu, 2026.

[85] Diabetes Prevention Program Research Group, "The Diabetes Prevention Program (DPP): Description of lifestyle intervention," *Diabetes Care*, vol. 25, no. 12, pp. 2165–2171, 2002. DOI: 10.2337/diacare.25.12.2165

[86] Look AHEAD Research Group et al., "The Look AHEAD study: A description of the lifestyle intervention and the evidence supporting it," *Obesity*, vol. 14, no. 5, pp. 737–752, 2006. DOI: 10.1038/oby.2006.84